MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.
Each subject is divided into 2 sections:
Section 1: Single Correct Type Questions (4 options, only one correct)
Section 2: Numerical Value Type Questions (answer to be entered as numerical value)
Marking Scheme:
Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.
Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

An electron moves with speed $2 \times 10^5 \text{ m/s}$ along the positive x-direction in the presence of a magnetic induction $B = \hat{i} + 4\hat{j} - 3\hat{k}$ (in tesla). The magnitude of the force experienced by the electron in newton is (charge on the electron $= 1.6 \times 10^{-19}\text{C}$)

(A) 1.18×10^{-13}

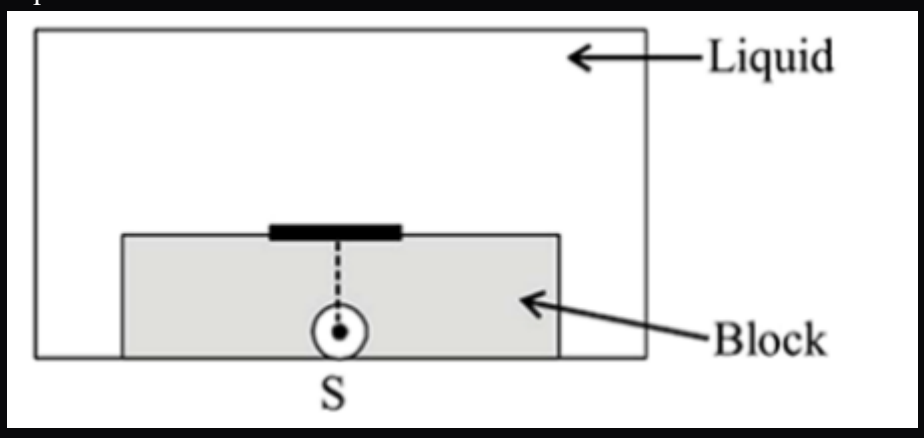
(B) 1.28×10^{-13}

(C) 1.6×10^{-13}

(D) 1.72×10^{-13}

Q.2

A point source S is placed at the bottom of a transparent block of height 10 mm and refractive index 2.72. It is immersed in a lower refractive index liquid as shown in the figure. It is found that the light emerging from the block to the liquid forms a circular bright spot of diameter 11.54 mm on the top of the block. The refractive index of the liquid is



(A) 1.21

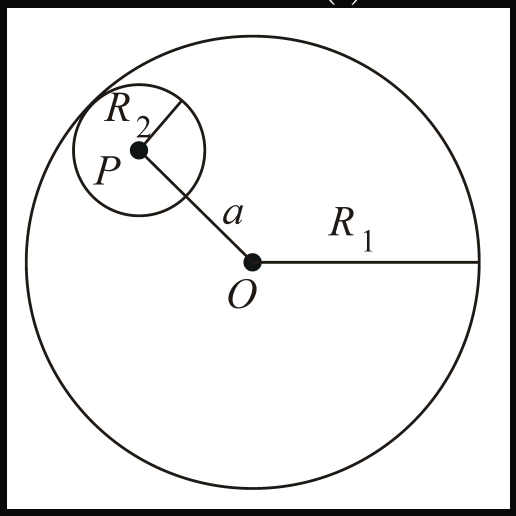
(B) 1.3

(C) 1.36

(D) 1.42

Q.3

Consider a uniform spherical charge distribution of radius R_1 centered at the origin O. In this distribution, a spherical cavity of radius, centered at P with distance $OP = a = R_1 - R_2$ (see figure) is made. If the electric field inside the cavity at position $\vec{r}$ is $\vec{E}\left(\vec{r}\right)$, then the correct statement is / are:



(A) $\vec{\mathrm{E}}$ is uniform, its magnitude is independent of R_2 , but its direction depends upon $\vec{\mathrm{r}}$

(B) $\vec{\mathrm{E}}$ is uniform, its magnitude is depends on R_2 , but its direction depends upon $\vec{\mathrm{r}}$

(C) $\vec{\mathrm{E}}$ is uniform, its magnitude is independent of a , but its direction depends upon $\vec{a}$

(D) $\vec{\mathrm{E}}$ is uniform, both its magnitude and direction depends upon $\vec{a}$

Q.4

Which of the following statements are incorrect?
(i)If there were no friction, Work need to be done to move a body up an inclined plane is zero.
(ii)If there were no friction, moving vehicles could not be stopped even by locking the brakes.
(iii)As the angle of inclination is increased, the normal reaction on the body placed on it increases.
(iv)A duster weighing 0.5 kg is pressed against a vertical board with a force of 11 N. If the coefficient of friction is 0.5, the work done in rubbing it upward through a distance of 10 cm is 0.55J.

(A) (i)and(ii)

(B) (i),(ii),(iv)

(C) (i),(iii),and(iv)

(D) All of these

Q.5

A body is executing simple harmonic motion. At a displacement x from mean position, its potential energy is $E_1 = 2J$ and at a displacement y from mean position, its potential energy is $E_2 = 8\ \mathrm{J}$. The potential energy E at a displacement $(x + y)$ from mean position is

(A) $10\ \text{J}$

(B) $14\ \text{J}$

(C) $18\ \text{J}$

(D) $4\ \text{J}$

Q.6

A body weighs $700\ \text{gm wt}$ on the surface of the earth. How much will it weigh on the surface of a planet whose mass is $\frac{1}{7}$ and radius is half that of the earth?

(A) $200\ \text{gm wt}$

(B) $400\ \text{gm wt}$

(C) $50\ \text{gm Wt}$

(D) $300\ \text{gm wt}$

Q.7

Assertion: An electric field is preferred in comparison to magnetic field for detecting the electron beam in a television picture tube.
Reason: Electric field require high voltage.

(A) If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

(B) If both Assertion and Reason are true but Reason is not correct explanation of the Assertion.

(C) If Assertion is true but the Reason is false.

(D) If Assertion is false but Reason is true.

Q.8

An open organ pipe of length l is sounded together with another organ pipe of length $l + x$ in their respective fundamental tones (given $x << l$). The beat frequency heard will be ?
(speed of sound is v)

(A) $\frac{vx}{4l^2}$

(B) $\frac{vx}{2l^2}$

(C) $\frac{vl^2}{2x}$

(D) $\frac{vx^2}{2l}$

Q.9

The electron in a H-atom makes transition from M shell to K shell. The ratio of magnitude of initial to final acceleration of the electron is

(A) $1:81$

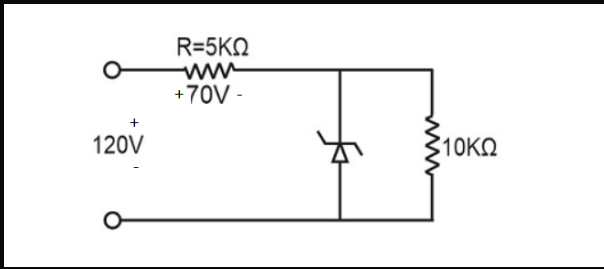
(B) $1:27$

(C) $27 : 1$

(D) $1 : 9$

Q.10

For the circuit shown in the figure :



(A) Current through Zener diode is $4\ \text{mA}$

(B) Current through Zener diode is $9\ \text{mA}$

(C) The output voltage is $60\ \text{V}$

(D) The output voltage is $40\ \text{V}$

Q.11

An ideal gas is heated at constant pressure and absorbs amount of heat Q. If the adiabatic exponent is γ then the fraction of change in the internal energy and performing the work, is

(A) $\gamma - 1$

(B) $1 - \frac{1}{\gamma}$

(C) $\frac{1}{\gamma - 1}$

(D) $\frac{1}{\gamma} - 1$

Q.12

Let $\lambda_\alpha, \lambda_\beta$ and λ'_α denote the wavelengths of the X-rays of the K_α, K_β and L_α lines in the characteristic X -rays for a metal. Then,

(A) $\frac{1}{\lambda_\beta} = \frac{1}{\lambda_\alpha} + \frac{1}{\lambda'_\alpha}$

(B) $\frac{1}{\lambda'_\alpha} = \frac{1}{\lambda_\beta} + \frac{1_\alpha}{\lambda_\alpha}$

(C) $\frac{1}{\lambda_\alpha} = \frac{1}{\lambda_\beta} + \frac{1}{\lambda' \alpha}$

(D) $\lambda_\alpha = \lambda_\beta + \lambda'_\alpha$

Q.13

When an ideal gas with pressure P and volume V is compressed isothermally to one-fourth of its volume, the pressure is P_1. When the same gas is compressed polytropically according to the equation $PV^{1.5}$ = constant to one-fourth of its initial volume, the pressure is P_2 The ratio $\frac{P_1}{P_2}$ is

(A) $\frac{1}{2}$

(B) $\frac{1}{2^{1.5}}$

(C) 2

(D) $2^{1.5}$

Q.14

Statement I If two non-ideal, unidentical batteries are connected in parallel with positive terminals are connected together, then the equivalent emf is smaller than either of two emf's.
Statement II The equivalent resistance is smaller than either of the two internal resistances.

(A) Both Statement$I$ and Statement II are true and the Statement II is the correct explanation of the Statement I

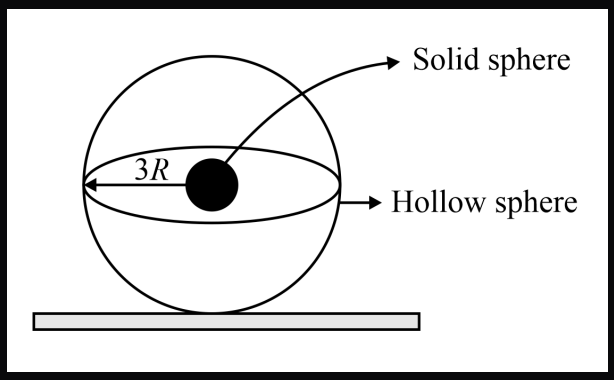
(B) Both Statement I and Statement II are true but the Statement II is not the correct explanation of the Statement I

(C) Statement I is false but Statement II is true.

(D) Both Statement Both I and Statement II are false.

Q.15

A solid ball of mass m and radius R is released from the position shown in a large hollow fixed shell of same mass m and radius $3R$ as shown in the figure. The displacement of the centre of mass of the system from its initial position when the solid ball touches the lower surface of the hollow shell is (centres of both the spheres coincide initially):



(A) 0

(B) $3\ R$

(C) $2\ R$

(D) R

Q.16

Using the principle of homogeneity of dimensions, which of the following is correct?

(A) $T^2 = \frac{4\pi^2 r^3}{GM}$

(B) $T^2 = 4\pi^2 r^3$

(C) $T^2 = \frac{4\pi^2 r^3}{G}$

(D) $T = \frac{2\pi^2 r^3}{G}$

Q.17

Two bars A and B of circular cross-section and of same volume and made of the same material are subjected to tension. If the diameter of A is half that of B and if the force applied to both the rods is the same and it is in the elastic limit, the ratio of extension of A to that of B will be

(A) $16 : 1$

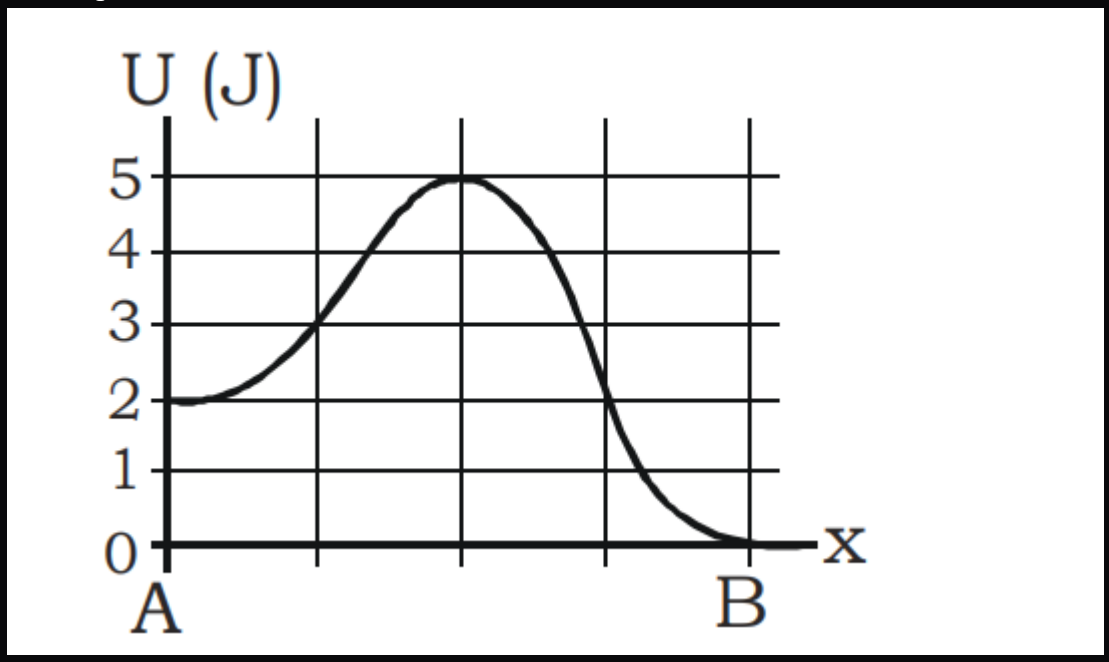
(B) $8 : 1$

(C) $4 : 1$

(D) $2 : 1$

Q.18

What minimum speed does a $100\ \text{g}$ particle need at point B to reach point A ? The graph shows potential energy versus position.



(A) $\sqrt{40}\ \text{m/s}$

(B) $\sqrt{60}\ \text{m/s}$

(C) $10\ \text{m/s}$

(D) $5\ \text{m/s}$

Q.19

The power of biconvex lens is 10 dioptre and the radius of curvature of each surface is $10\ \text{cm}$. Then the refractive index of the material of the lens is :

(A) $\frac{3}{2}$

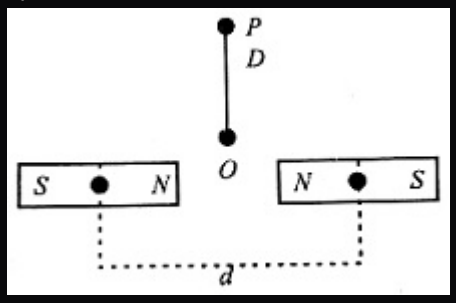
(B) $\frac{4}{3}$

(C) $\frac{9}{8}$

(D) $\frac{5}{3}$

Q.20

Two identical bar magnets are fixed with their centres at a distance d apart. A stationary charge Q is placed at P in between the gap of the two magnets at a distance D from the centre O as shown in the figure. The force on the charge Q is



(A) Directed perpendicular to the plane of paper

(B) Zero

(C) Directed along OP

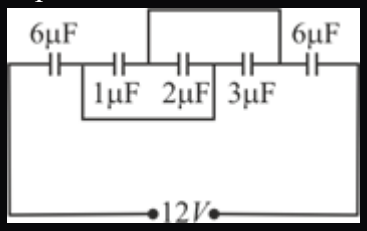
(D) Directed along PO

Q.21

Two identical discs are positioned on a vertical axis. The bottom disc is rotating with angular velocity ω_0. The top disc is initially at rest. It is allowed to fall and sticks to the lower disc. Ratio of Kinetic Energy before and after collision is $M:1$, find M.

Q.22

Figure shows five capacitors connected across a $12\ \mathrm{V}$ power supply. What is the potential drop across the $2\mu\mathrm{F}$ capacitor?



Q.23

A uniform copper rod 1 m long is insulated on its sides and has the ends exposed to ice and steam respectively. If there is a layer of water $2\ \mathrm{mm}$ thick at each end, then what is temperature gradient in the bar ? (Given: Thermal conductivity of copper is $435\ \mathrm{Wm^{-1}k^{-1}}$ and that of water is $0.435\ \mathrm{Wm^{-1}k^{-1}}$)

Q.24

Two nicol prisms A and B are placed in the path of a beam of unpolarised light, so that no light emerges out of B. In between these two, a third nicol C is placed such that its principal section is at an angle of 30° with that of A. The percentage of intensity of incident unpolarized light that emerges from B (round off answer to nearest integer)

Q.25

A heater coil is rated $100\ \mathrm{W}, 200\ \mathrm{V}$. It is cut into two identical parts. Both parts are connected together in parallel, to the same source of $200\ \mathrm{V}$. Calculate the energy liberated (in watt) in the new combination

Q.26

Deficiency of which vitamin causes Pernicious anaemia?

(A) Vitamin K

(B) Vitamin A

(C) Vitamin C

(D) Vitamin B_{12}

Q.27

Which of the following is the best way in preventing the rusting of iron?

(A) Making it a cathode.

(B) Placing it in saline water.

(C) Both (a) and (b).

(D) None of these.

Q.28

Given:
(a) $n = 5, m_l = 0$
(b) $n = 4, m_l = +2, m_s = -1/2$
The maximum number of electron(s) in an atom that can have the quantum numbers as given in (a) and (b) are respectively:

(A) 8 and 1

(B) 10 and 2

(C) 10 and 4

(D) 25 and 1

Q.29

Which of the following statement is incorrect about the collision theory of chemical reaction?

(A) It considers reacting molecules or atoms to be hard spheres and ignores their structural features.

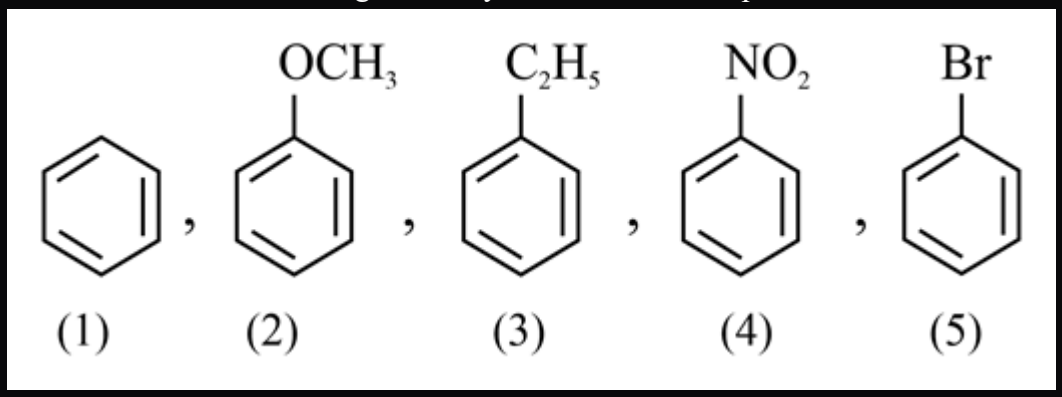
(B) Number of effective collisions determines the rate of reaction.

(C) Colliision of atoms or molecules possessing sufficient threshold energy results into the product formation.

(D) Molecules should collide with sufficient threshold energy and proper orientation for the collision to be effective.

Q.30

Give the order of decreasing reactivity towards an electrophile.



(A) $2 > 3 > 1 > 5 > 4$

(B) $3 > 1 > 2 > 5 > 4$

(C) $5 > 4 > 2 > 3 > 1$

(D) $1 > 5 > 2 > 3 > 4$

Q.31

Reduction of hexose A (molecular formula $\mathrm{C_6H_{12}O_6}$) with sodium borohydride gives compound B and C. Compound B is optically inactive, whereas compound C is optically active. Which of the following is compound A?

(A) D-fructose

(B) D-glucose

(C) D-mannose

(D) D-galactose

Q.32

$10mL$ of $0.2\mathrm{M}$ solution of compound, $\mathrm{Na_2CO_3 \cdot NaHCO_3 \cdot 2H_2O}$ is titrated against $0.05\mathrm{M\ HCl}$, $y\ \mathrm{mL}$ of HCl is used when phenolphthalein is used as an indicator and $x\ \mathrm{mL}$ of HCl is used when methyl orange is the indicator in two separate titrations. The value of $(x - y)$ will be

(A) $120\ \mathrm{mL}$

(B) $80\ \mathrm{mL}$

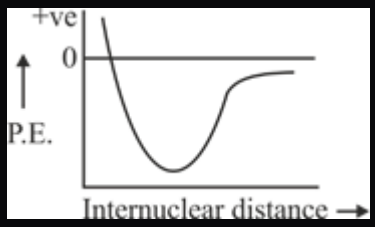
(C) $100\ \mathrm{mL}$

(D) $40\ \mathrm{mL}$

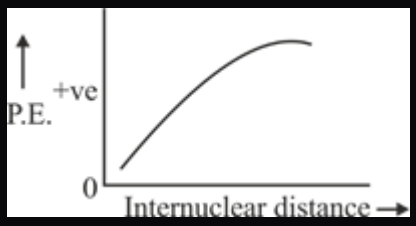
Q.33

Which of the following will refer to the potential energy diagram, when two helium atoms approach each other for the formation of a chemical bond?

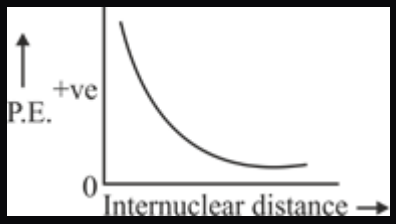
(A)



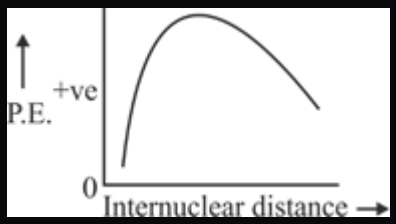
(B)



(C)



(D)



Q.34

Identify the correct statement for change of Gibb's energy for a system $(\Delta G_{\text{system}})$ at constant temperature and pressure

(A) If $\Delta G_{\text{system}} = 0$, the system has attained equilibrium

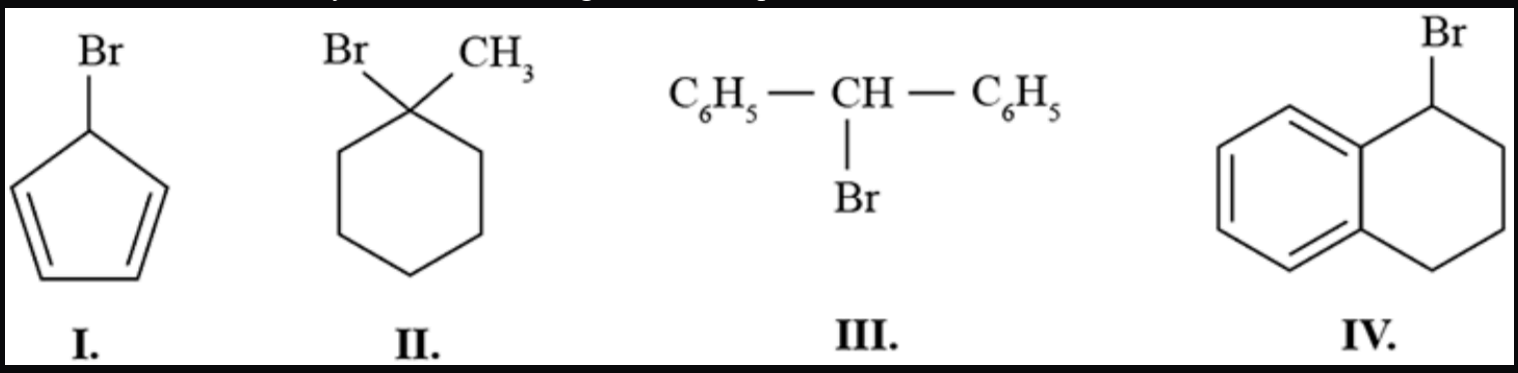
(B) If $\Delta G_{\text{system}} = 0$, the system is still moving in a particular direction

(C) If $\Delta G_{\text{system}} < 0$, the process is not spontaneous

(D) If $\Delta G_{\text{system}} > 0$, the process is not spontaneous

Q.35

Consider the S_N1 solvolysis of the following halides in aqueous formic acid?



Which one of the following is the correct sequence of the halides given above in the decreasing order of their reactivity?

(A) $\mathrm{III} > \mathrm{IV} > \mathrm{II} > \mathrm{I}$

(B) $\mathrm{II} > \mathrm{IV} > \mathrm{I} > \mathrm{III}$

(C) $\mathrm{I} > \mathrm{II} > \mathrm{III} > \mathrm{IV}$

(D) $\mathrm{III} > \mathrm{I} > \mathrm{II} > \mathrm{IV}$

Q.36

Which of the following will undergo oxidation reaction with $\mathrm{MnO_2}$?

(A) Hex-3-ene-1-ol

(B) Cyclohex-3-ene-1-ol

(C) Hex-2-ene-1-ol

(D) All of these

Q.37

The end product in the below reaction is

$\mathrm{C_2H_5NH_2} \xrightarrow{HNO_2} A \xrightarrow{PCl_5} B \xrightarrow{NH_3} C$

(A) Ethyl cynide

(B) Ethyl amine

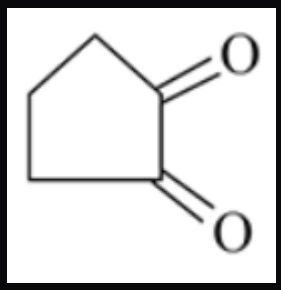
(C) Methyl amine

(D) Acetamide

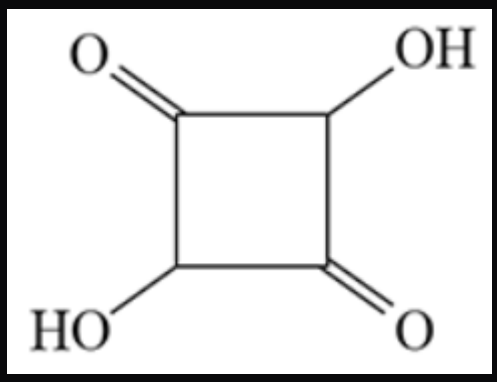
Q.38

Compound (A) $\xrightarrow{\text{2 mol of } \mathrm{HIO_4}}$ two moles of glyoxalic acid. The compound ' A ' is

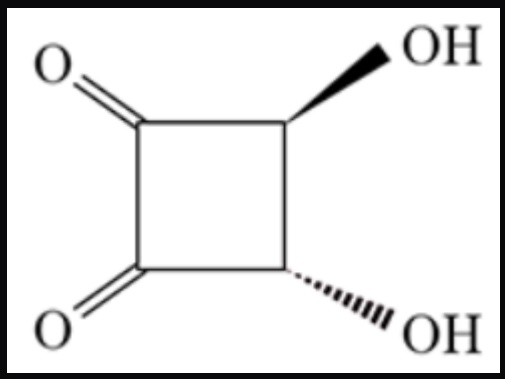
(A)



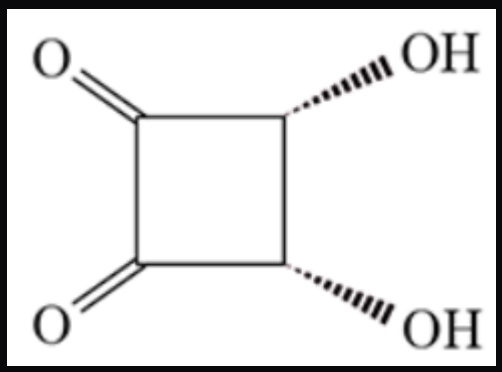
(B)



(C)



(D)



Q.39

Select the set having incorrect statements given here.

(i) Manganese exhibits $+7$ oxidation state

(ii) Zinc forms coloured ions

(iii) $[CoF_6]^{3-}$ is diamagnetic

(iv) Sc forms $+4$ oxidation state

(v) Zn exhibits only $+2$ oxidation state

(A) (i), (ii) and (iii)

(B) (ii), (iii) and (iv)

(C) (i), (iii) and (iv)

(D) (i), (iii) and (v)

Q.40

The CFSE for $[CoCl_6]^{4-}$ complex is 18000 cm^{-1} . The Δ for $[CoCl_4]^{2-}$ will be

(A) $9000\ cm^{-1}$

(B) $4000\ cm^{-1}$

(C) $8000\ cm^{-1}$

(D) $2000\ cm^{-1}$

Q.41

Which one of the following on treatment with 50% aqueous sodium hydroxide yields the corresponding alcohol and acid?

(A) C_6H_5CHO

(B) $CH_3CH_2CH_2CHO$

(C)

$$CH_3-\overset{\overset{O}{||}}{C}-CH_3$$

(D) $C_6H_5CH_2CHO$

Q.42

$Mg(OH)_2$ is precipitated when NaOH is added to a solution of Mg^{2+}. If the final concentration of Mg^{2+} is 10^{-10} M , the concentration of $OH^-(M)$ in the solution is
[Solubility product for $Mg(OH)_2 = 5.6 \times 10^{-12}$]

(A) 0.056

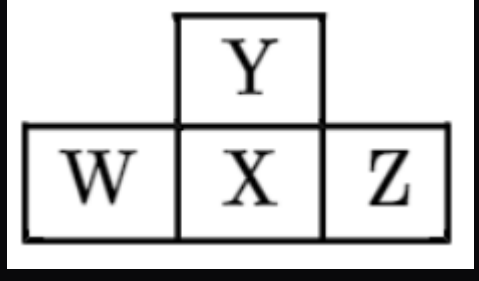
(B) 0.12

(C) 0.24

(D) 0.025

Q.43

Consider the following four elements, which are represented according to long form of periodic table



Here W, Y and Z are left, up and right elements with respect to the element 'X' and 'X' belongs to 16th group and 3rd period. Then according to given information the incorrect statement regarding given elements is

(A) Maximum electronegativity: Y

(B) Maximum catenation property: X

(C) Maximum electron affinity: Z

(D) Y exhibits variable covalency

Q.44

Assertion (A) : Noble gases have very low boiling points.
Reason (R) : All noble gases have general electronic configuration of $ns^2\ np^6$ (except He)

(A) Both (A) and (R) are true but (R) is correct explanation of (A)

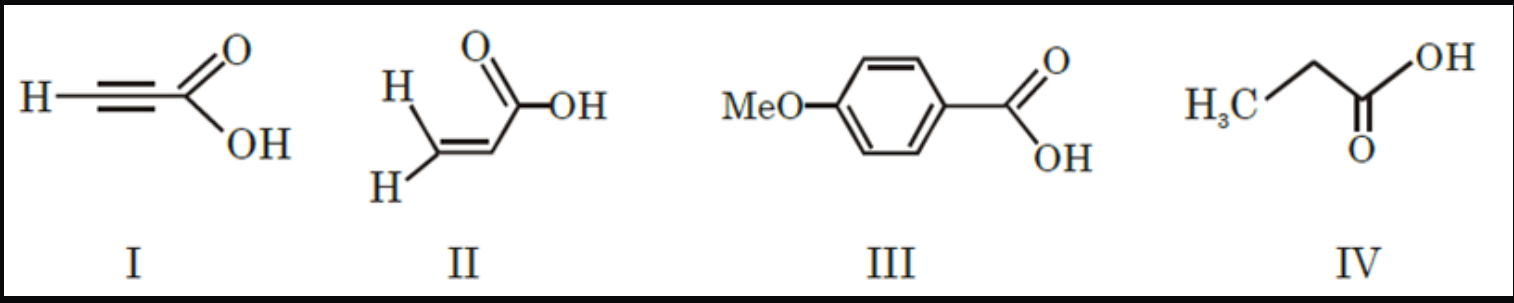
(B) (A) is false but (R) is true

(C) (A) is true but (R) is false

(D) Both (A) and (R) are true but (R) is not the correct explanation of (A)

Q.45

The correct order of acid strength of the following carboxylic acids is -



(A) $I > III > II > IV$

(B) $III > II > I > IV$

(C) $II > I > IV > III$

(D) $I > II > III > IV$

Q.46

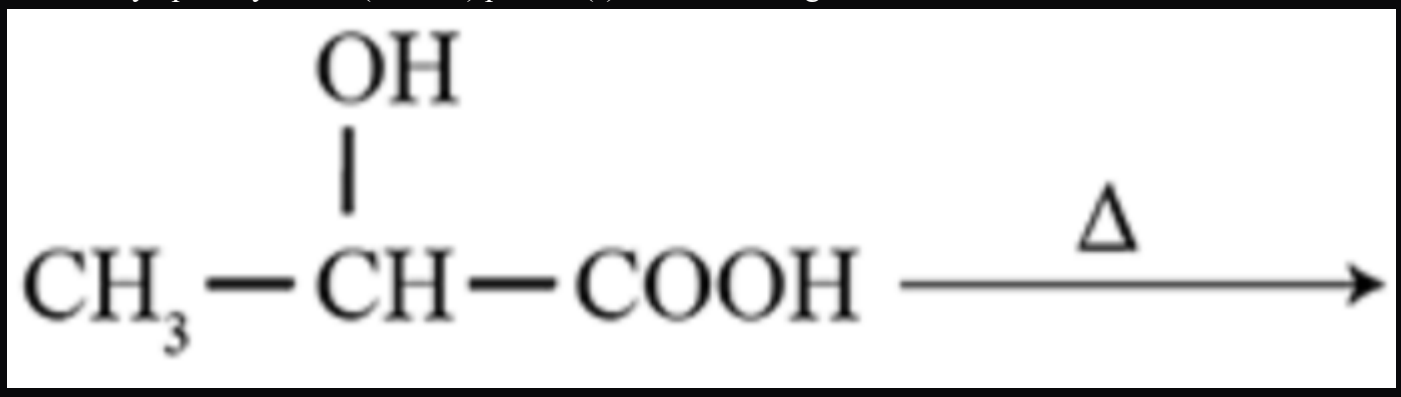
The equivalent conductance of $1M$ benzoic acid is $12.8 ohm^{-1}\ cm^2$ and if the conductance of benzoate ion and H^+ ion at infinite dilution are 42 and $288.42 ohm^{-1}\ cm^2$ respectively. Calculate percentage of degree of dissociation. Report your answer by rounding upto nearest whole number.

Q.47

0.14 gram of $NaHCO_3$ tablet is added to neutralise 5 ml of 0.1 N HCl produced in stomach of a man. Then percentage purity of $NaHCO_3$ tablet is -

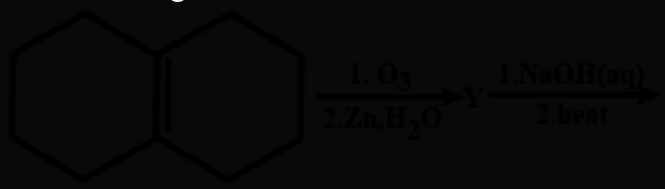
Q.48

How many optically active (Lactide) product(s) are formed in given reaction?



Q.49

In the scheme given below, the total number of intramolecular Aldol condensation products formed from 'Y' is:



Q.50

The gas phase decomposition of dimethylether follows first order kinetics

$CH_3 - O - CH_3(g) \rightarrow CH_4(g) + H_2(g) + CO_{(g)}$

The reaction is carried out in constant volume container at $500^{\circ}C$ and has a half-life of 14.5 min. Initially only dimethylether is present at a pressure of 0.40 atm. The total pressure of the system after 12 min is $\frac{x}{100}$ atm. The value of x is

[Given $10^{0.25} = 1.778$]

Q.51

If $\vec{a} = x\hat{i} + y\hat{j} + z\hat{k}, \vec{b} = \hat{i} + \hat{j} + \hat{k}$ where $x, y, z \in \{0, 3, -3, 6, -6\}$, then the number of non-zero vectors $\vec{a}$ such that $\vec{a} \cdot \vec{b} = 0$ are

(A) 20

(B) 18

(C) 15

(D) 12

Q.52

Let $f(x) = x^{2010} + x^{1010} - x^{510} + x^{210} + x^2$. If $f(x)$ is divided by $x^2\left(x^2 - 1\right)$, then we get remainder as $g(x)$, function of x. If $g(x)$ is defined from R to R, then:

(A) $g(x)$ is injective but not surjective

(B) $g(x)$ is surjective but not injective

(C) $g(x)$ is neither injective nor surjective

(D) $g(x)$ is injective and surjective

Q.53

The range of the function $f(x)=\sin^{-1}\left[x^2-\frac{1}{3}\right]-\cos^{-1}\left[x^2+\frac{2}{3}\right]$ is (where, $[x]$ represents the greatest integer value of x)

(A) $[-\pi, 0]$

(B) $\{-\pi, 0\}$

(C) $\{0, \pi\}$

(D) $\{0, \pi, -\pi\}$

Q.54

A data consists of n observations: $x_1, x_2, \ldots, x_n$. If $\sum_{i=1}^{n}\left(x_i+1\right)^2=11n$ and $\sum_{i=1}^{n}\left(x_i-1\right)^2=7n$, then the variance of this data is

(A) 5

(B) 8

(C) 6

(D) 7

Q.55

The solution of the differential equation $\frac{dy}{dx}+\frac{xy}{1-x^2}=x\sqrt{y},(|x|<1)$ is $\sqrt{y}=-\frac{f(x)}{3}+C\left(1-x^2\right)^{\frac{1}{4}}$, where $f\left(\frac{1}{2}\right)=\frac{3}{4}$ and C is an arbitrary constant. Then, the value of $f\left(-\frac{1}{2}\right)$ is

(A) $-\frac{3}{4}$

(B) $\frac{3}{4}$

(C) $\frac{1}{4}$

(D) $\frac{3}{2}$

Q.56

Let O be an interior point of triangle ABC, such that
$2\overrightarrow{OA}+3\overrightarrow{OB}+4\overrightarrow{OC}=0$, then the ratio of the area of ΔABC to the area of ΔAOC is

(A) $3:1$

(B) $3:2$

(C) $2:1$

(D) $4:3$

Q.57

The derivative of $\cos\left(2\tan^{-1}\left(\sqrt{\frac{1-x}{1+x}}\right)\right)-2\cos^{-1}\left(\sqrt{\frac{1-x}{2}}\right)$ W. r. to x is

(A) $1-\frac{1}{\sqrt{1-x^2}}$

(B) $1-\frac{1}{\sqrt{1+x^2}}$

(C) $2-\frac{1}{\sqrt{1-x^2}}$

(D) $2-\frac{1}{\sqrt{1+x^2}}$

Q.58

The area bounded by $y=f(x),\ x=1,\ x=4$ and the x-axis is 12 square units where $f(x)$ is a differentiable function such that $f(1)=3$ and $f(4)=8$. If $f(x)$ is a bijective function which is always positive, then the area (in square units) bounded by $y=f^{-1}(x),\ x=3,\ x=8$ and the x-axis is equal to

(A) 29

(B) 32

(C) 12

(D) 17

Q.59

For any three positive real numbers a, b and c, $4\left(9c^2+b^2\right)+9\left(a^2-2ac\right)=6b(2c+a)$. Then:

(A) b, c and a are in A.P.

(B) c, a and b are in A.P.

(C) a, b and c are in G.P.

(D) b, c and a are in G.P.

Q.60

India plays a series of 5 cricket matches against Australia with $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{6}$ chances of winning, loosing and drawing each match. Result of one match is independent of the other. The probability of India winning a series with margin of 4 wins or more is: India cannot win 4 times and draw 1 time.

(A) $\frac{7}{3(2^6)}$

(B) $\frac{11}{3(2^6)}$

(C) $\frac{1}{12}$

(D) $\frac{10}{3(2^6)}$

Q.61

Given real numbers x, y, z, such that $x+y+z=3, x^2+y^2+z^2=5, x^3+y^3+z^3=7$, find $x^4+y^4+z^4$.

(A) 18

(B) 16

(C) 9

(D) 24

Q.62

The value of a in order that $f(x)=\sin x-\cos x-ax+b$ decreases for all real values is given by

(A) $a \geq \sqrt{2}$

(B) $a<\sqrt{2}$

(C) $a \geq 1$

(D) $a<1$

Q.63

In a set of real numbers a relation R is defined as xRy such that $|\mathrm{x}|+|\mathrm{y}| \leq \frac{1}{2}$, then relation R is

(A) reflexive and symmetric but not transitive

(B) symmetric but not transitive and reflexive

(C) transitive but not symmetric and reflexive

(D) none of reflexive, symmetric and transitive

Q.64

If $(1!)^2+(2!)^2+(3!)^2+\cdots+(99!)^2+(100!)^2$ is divided by 100 , then the remainder obtained is

(A) 27

(B) 28

(C) 17

(D) 14

Q.65

P is the point of concurrency of three straight lines L_1, L_2, and L_3 which are mutually perpendicular to each other. If one of them intersects the x-axis and other intersects the y-axis, while the third line passes through a fixed point (0, 0, c) on the z-axis, then which of the following options satisfy the locus of the point P ?

(A) (0, 0, c)

(B) (c, c, 0)

(C) (0, c, c)

(D) (0, c, 0)

Q.66

A is a (3×3) diagonal matrix having integral entries such that $\det(A) = 120$, number of such matrices is $10n$ Then n is

(A) 24

(B) 36

(C) 16

(D) 48

Q.67

If e_1 is the eccentricity of the ellipse $\frac{x^2}{16} + \frac{y^2}{25} = 1$ and e_2 is the eccentricity of the hyperbola passing through the foci of the ellipse and $e_1 e_2 = 1$, then the equation of the hyperbola is

(A) $\frac{x^2}{9} - \frac{y^2}{16} = 1$

(B) $\frac{x^2}{16} - \frac{y^2}{9} = -1$

(C) $\frac{x^2}{9} - \frac{y^2}{25} = 1$

(D) None of these

Q.68

Let $\mathrm{A}\left(\mathrm{z}_1\right), \mathrm{B}\left(\mathrm{z}_2\right)$ and $\mathrm{C}\left(\mathrm{z}_3\right)$ lie on the circle $|z - i| = 1$ and satisfy the equation $3\mathrm{z}_1 + \mathrm{i} = 2\mathrm{z}_2 + 2\mathrm{z}_3$. If D is the centre of the circle $|z - i| = 1$, then area of quadrilateral ABCD equals :-

(A) $\frac{\sqrt{3}}{4}$

(B) $\frac{\sqrt{5}}{4}$

(C) $\frac{\sqrt{11}}{4}$

(D) $\frac{\sqrt{7}}{4}$

Q.69

Let $I_1 = \int_0^{\alpha} \frac{1+2\cos x}{1+e^x} dx$ and $I_2 = \int_0^{\alpha} \frac{1+e^x}{1+2\cos x} dx$, where α is the root of the equation $2\cos x - e^x = 0$. Then,

(A) $I_1 = I_2$

(B) $I_1 > I_2$

(C) $I_1 + I_2 = 0$

(D) $I_1 < I_2$

Q.70

The system of simultaneous equations $kx + 2y - z = 1, (k-1)y - 2z = 2$ and $(k+2)z = 3$ has a unique solution, if k equals

(A) -2

(B) -1

(C) 0

(D) 1

Q.71

Consider a polynomial $P(x)$ of the least degree that has a maximum equal to 6 at $x = 1$ and a minimum equal to 2 at $x = 3$. Then, the value of $P(2) + P'(0) - 7$, is

Q.72

The number of 6-digit numbers, such that the digits of each number are all from the set $\{1, 2, 3, 4, 5\}$;and any digit that appears in the number appears at least twice; are equal to (Example : 225252 is an admissible number, while 222133 is not)

Q.73

The extremities of the diagonal of a rectangle are $(-4, 4)$ and $(6, -1)$. A circle cirumscribes the rectangle and cuts intercept of length AB on the y-axis. The length of AB is

Q.74

Let $A = \{1, 2, 3, 4, 5, 6, 7, 8\}, B = \{9, 10, 11, 12\ 13, 14, 15, 16\}$ and $C = \{17, 18, 19, 20, 21, 22, 23, 24\}$. Find the number of triples (x, y, z) such that $x \in A, y \in B, z \in C$ and $x + y + z = 36$.

Q.75

The locus of the middle points of the focal chords of parabola, $y^2 = 8x$ is another parabola whose length of latus rectum is

Answers & Solutions

Q.1 Answer:

1.6×10^{-13}

Solution:

$\vec{v} = 2 \times 10^5 \hat{i}$ **and** $\vec{B} = (\hat{i} + 4\hat{j} - 3\hat{k})$

$\vec{F} = q(\vec{v} \times \vec{B}) = -1.6 \times 10^{-19} \left[2 \times 10^5 \hat{i} \times (i + 4\hat{j} - 3\hat{k})\right]$

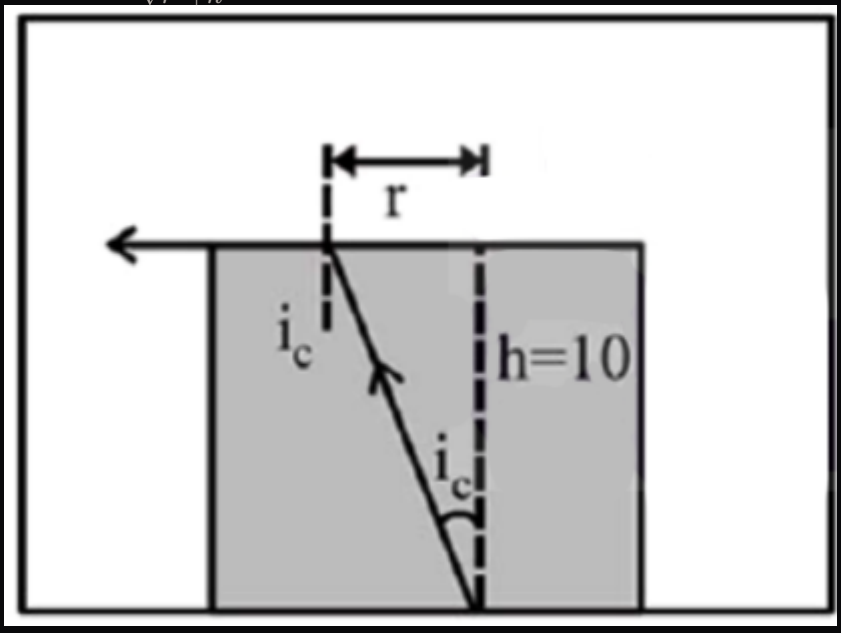
$= -1.6 \times 10^{-19} \times 2 \times 10^5 [\hat{i} \times \hat{i} + 4(\hat{i} \times \hat{j}) - 3(\hat{i} \times \hat{k})]$
$= -3.2 \times 10^{-14} [0 + 4\hat{k} + 3\hat{j}] = 3.2 \times 10^{-14} (-4\hat{k} - 3\hat{k})$
$\Rightarrow |\vec{F}| = 3.2 \times 10^{-14} \times 5 = 1.6 \times 10^{-13} N.$

Q.2 Answer:

1.36

Solution:

$\sin i_c = \frac{r}{\sqrt{r^2+h^2}}$



$\Rightarrow \frac{n_l}{n_B} = \frac{r}{\sqrt{r^2+h^2}} \times 2.72$
$\Rightarrow n_l = \frac{r}{\sqrt{r^2+h^2}} \times 2.72 = \frac{5.77}{11.54} \times 2.72 = 1.36$

Q.3 Answer:

$\vec{E}$ is uniform, both its magnitude and direction depends upon $\vec{a}$

Solution:

The Electric field inside the given cavity can be written as : $\vec{E} = \frac{\rho \vec{OP}}{3\varepsilon_0}$

Which can be written as :

$\vec{OP} = \vec{a} = \vec{R_1} - \vec{R_2}$

$\Rightarrow \vec{E} = \frac{\rho \vec{a}}{3\varepsilon_0}$, **here** ρ **is the charge density and** ε_0 **is the permittivity of free space, which is constant.**

Q.4 Answer:

(i),(iii),and(iv)

Solution:

The explanation are given below

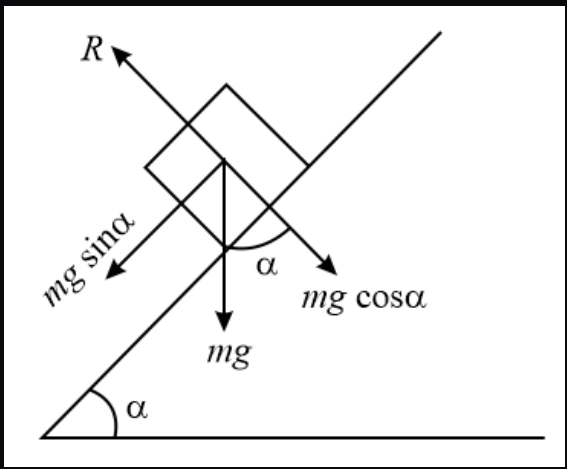
(i)If a body is moved up in inclined plane, then the work done against friction force is zero as there is no friction. But a work has to be done against the gravity. So, this statement is incorrect.

(ii)If there were no friction, moving vehicles could not be stopped by locking the brakes. Vehicles are

stopped by air friction only.

So , This Statement is correct.

(iii) In this situation the normal reaction is given by

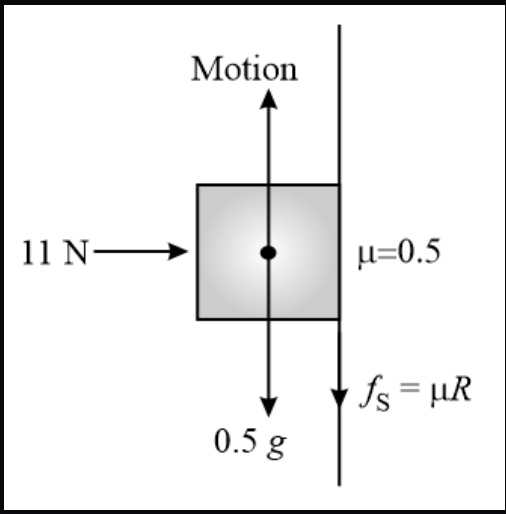


$R = mg\cos\alpha \qquad .....(i)$

If α increase then the value of cos α also decreases.

So, this Statement is incorrect.

(iv)When the duster is rubbing upward then an external force is applied and its value is



F'**=0.5g+**μR

$\therefore \quad F'' = 0.5g + 0.5 \times 11$

Or $F'' = (0.5 \times 10 + 5.5)$**N (Here R=11 N)**

Or $F'' = 10.5$N

Hence, work done in rubbing the duster through a distance of 10 cm.

$W = F'' \times d$

$\Rightarrow \qquad W = 10.5 \times \frac{10}{100}$J

Or $F'' = 10.5$J

Q.5 Answer:

18 J

Solution:

$E_1 = \frac{1}{2}kx^2, E_2 = \frac{1}{2}ky^2$

$E = \frac{1}{2}k(x+y)^2 = \frac{1}{2}kx^2 + \frac{1}{2}ky^2 + kxy = E_1 + E_2 + 2\sqrt{E_1E_2} = 2 + 8 + 2\sqrt{16} = 18J$

Q.6 Answer:

400 gm wt

Solution:

The formula of weight is,

$w' = mg'$ **but**

$g' = \frac{GM'}{R'^2}$

$\therefore\ g' = \frac{G \times \frac{M}{7}}{\left(\frac{R}{2}\right)^2} = \frac{4GM}{7R^2} = \frac{4}{7}g$

$w' = mg' = 700 \times g' = 700 \times \frac{4}{7} = 400$ gm wt

Q.7 Answer:

If Assertion is false but Reason is true.

Solution:

A - False, R-True.

If electric field is used for detecting the electron beam, then very high voltage will have to be applied or very long tube will have to be taken.

Q.8 Answer:

$\frac{vx}{2l^2}$

Solution:

$l = \frac{\lambda_1}{2} \therefore \lambda_1 \Rightarrow 2l$

$\therefore f_1 = \frac{v}{2l}$

Similarly $f_2 = \frac{v}{2l\left(1+\frac{x}{l}\right)} = \frac{v}{2l}\left(1 - \frac{x}{l}\right)$

$\therefore \Delta f = f_1 - f_2 = \frac{v}{2l}\left(1 - 1 + \frac{x}{l}\right) = \frac{vx}{2l^2}$

Q.9 Answer:

1 : 81

Solution:

Now, centripetal acceleration is given by

$a = \omega^2 r = \frac{v^2}{r}$

Now, $v \propto \frac{1}{n}$ **and** $r \propto n^2$

[according to Bohr's model]

So, $\frac{v^2}{r} \propto \frac{1}{n^4}$ **or** $a \propto \frac{1}{n^4}$

As m **shell corresponds to** $n = 3$ **and** K **shell correspond to** $n = 1$

So, $\frac{a_m}{a_k} = \left(\frac{1}{3}\right)^4 = \frac{1}{81} = 1 : 81$

Q.10 Answer:

Current through Zener diode is 9 mA

Solution:

Here, $I = \frac{70}{5000} = 1.4 \times 10^{-2}\ \text{A}$

$V_{\text{Z}} = 120 - 70$

$= 50\ \text{V}$ **(output voltage Across diode)**

$I_{\text{L}} = \frac{50}{10000} = 5 \times 10^{-3}\ \text{A} = 0.5 \times 10^{-2}\ \text{A}$

$\Rightarrow \quad I_{\text{Z}} = (1.4 - 0.5) \times 10^{-2}\ \text{A}$

$= 9 \times 10^{-3}\ \text{A}$

$= 9\ \text{mA}$

Q.11 Answer:

$\frac{1}{\gamma - 1}$

Solution:

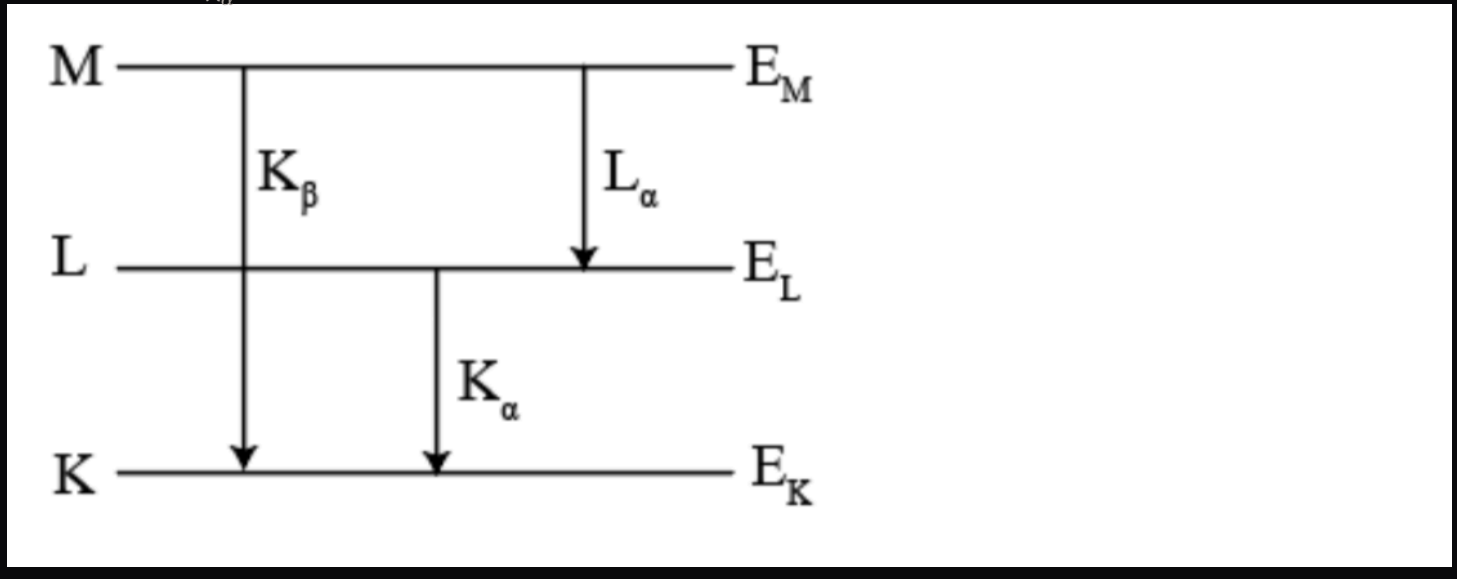
$$\Delta U = nC_V \Delta T = \frac{nR\Delta T}{(\gamma - 1)}$$
$$\omega = P \cdot \Delta V = nR\Delta T$$
$$\frac{\Delta U}{\omega} = \frac{1}{\gamma - 1}$$

Q.12 Answer:

$\frac{1}{\lambda_\beta} = \frac{1}{\lambda_\alpha} + \frac{1}{\lambda'_\alpha}$

Solution:

$\text{E}_{\text{L}} - \text{E}_{\text{K}} = \frac{\text{hc}}{\lambda_\alpha} \ldots (\text{i})$



$$E_M - E_K = \frac{hc}{\lambda_\beta} \dots \text{(ii)}$$

$$E_M - E_L = \frac{hc}{\lambda'_\alpha} \dots \text{(iii)}$$

$$\text{(ii)} - \text{(i)} \Rightarrow E_M - E_L = \frac{hc}{\lambda'_\alpha} = \frac{hc}{\lambda_\beta} - \frac{hc}{\lambda_\alpha}$$

$$\frac{1}{\lambda_\beta} = \frac{1}{\lambda_\alpha} + \frac{1}{\lambda'_\alpha}$$

Q.13 Answer:

$\frac{1}{2}$

Solution:

For isothermal process : PV = constant

or $\quad PV = P_1 \frac{V}{4} \quad \Rightarrow P_1 = 4P$

For polytropic process : PV $^{1.5}$ = constant

$$PV^{1.5} = P_2\left(\frac{V}{4}\right)^{1.5} \quad \Rightarrow \quad P_2 = \left(2^2\right)^{3/2} P = 8P$$

$\therefore \quad \frac{P_1}{P_2} = \frac{1}{2}$

Q.14 Answer:

Statement I is false but Statement II is true.

Solution:

$$E_{eq} = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2}}{1/r_1 + 1/r_2}$$

$$\Rightarrow E_{eq} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$$

$$\text{or } E_{eq} = E_1 \left[\frac{r_2 + r_1\,(E_2/E_1)}{r_1 + r_2}\right]$$

$\Rightarrow E_{eq} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$

or $E_{eq} = E_1 \left[\frac{r_2 + r_1 (E_2/E_1)}{r_1 + r_2}\right]$

So, E_{eq} can be greater than, less than or equal to E_1 depending upon the condition whether $E_2 > E_1, E_2 = E_1$ or $E_2 < E_1$

Also, $r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$ which is less than both of r_1, r_2

Q.15 Answer:

R

Solution:

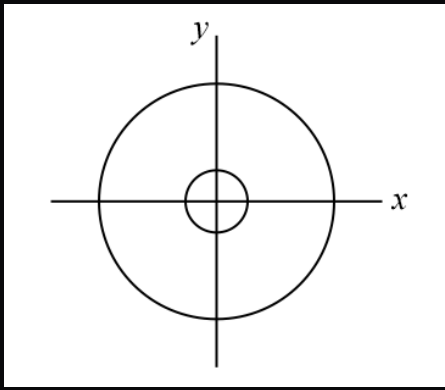
Initially coordinate of centre of mass of the ball and shell is $(0,\ 0)$**.**

When the solid ball touches the lower surface of hollow sphere, the centre of the solid ball gets shifted to a distance equal to $2R$**.**

Thus, the centre of mass of the system will be given as $y_{\text{cm}} = \frac{m(2\text{R}) + m \times 0}{m+m} = R$

So finally coordinate of COM **of the system is** $(0, -R)$**.**

The displacement of the centre of mass is $[0 - (-R)] = R$**.**



Q.16 Answer:

$T^2 = \frac{4\pi^2 r^3}{GM}$

Solution:

$T^2 = \frac{4\pi^2 r^3}{GM}$

Taking dimensions on both sides, we get

$[T]^2 = \frac{[L]^3}{[M^{-1}L^3T^{-2}M]} = \left[M^0L^0T^2\right]$

$\therefore$ **LHS** $=$ **RHS**

Now, $T^2 = 4\pi^2 r^2$

Taking dimensions on both sides

$[T]^2 = \left[L^2\right] \quad \therefore LHS \neq RHS$

Now, $T^2 = \frac{4\pi r^3}{G}$

Taking dimension on both sides

$[T]^2 = \frac{[L]^3}{[M^{-1}L^3T^2]} = \left[M^1L^0T^{-2}\right] \quad \therefore LHS \neq RHS$

Now, $\text{T} = \frac{4\pi^2 r^3}{G}$

$[T] = \frac{\left[L^3\right]}{[M^{-1}L^3T^2]} = \left[M^1L^0T^2\right] \therefore LHS \neq RHS$

Taking dimensions on both sides

Q.17 Answer:

16 : 1

Solution:

We have, $Y = \frac{F}{A} \times \frac{l}{\Delta l}$

and $V = Al$

or $l = \frac{V}{A}$

$\therefore Y=\frac{FV}{A^2\Delta l}$
$\Rightarrow \Delta l \propto \frac{1}{A^2}$
or $\Delta l \propto \frac{1}{D^4}$
$\Rightarrow \quad \frac{\Delta l_A}{\Delta l_B}=\frac{D_B^4}{D_A^4}=\frac{1^4}{\left(\frac{1}{2}\right)^4}=16$

Q.18 Answer:

$10 \mathrm{~m/s}$

Solution:

To reach at point A particle must cross the peak point.

Loss in KE **= gain in** PE

$$\frac{1}{2} \times \frac{100}{1000} \times \mathrm{v}^2=(5-0)$$
$$\mathrm{v}^2=100$$
$$\mathrm{v}=10 \mathrm{~m/s}$$

Q.19 Answer:

$\frac{3}{2}$

Solution:

$P(\text{ in dioptre })=\frac{100}{f(\text{ in cm})} \quad \therefore f=\frac{100}{10}=10 \text{ cm}$

According to lens maker's formula, $\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

For biconvex lens, $R_1=+R, R_2=-R$

$\therefore \quad \frac{1}{f}=(\mu-1)\left(\frac{1}{R}+\frac{1}{R}\right) ; \frac{1}{10}=(\mu-1)\left(\frac{2}{10}\right) \Rightarrow \quad \mu=\frac{1}{2}+1=\frac{3}{2}$

Q.20 Answer:

Zero

Solution:

Force on a charged particle is given by $F=qvB$ **Here** $v=0$ **and also resultant B is zero.**

$\therefore \text{Force}=0$

Q.21 Answer:

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Solution:

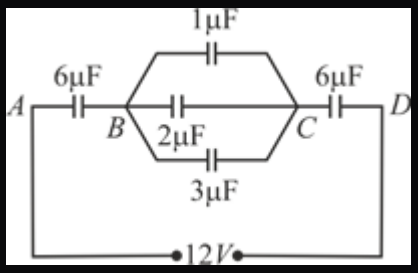
$I\omega_0=2I\omega(M.I.\text{ of one disc })$
$\therefore K=\frac{1}{2}2I\omega^2$
$=\frac{1}{2}\left(\frac{1}{2}I\omega_0^2\right)$

Q.22 Answer:

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Solution:

The equivalent figure is



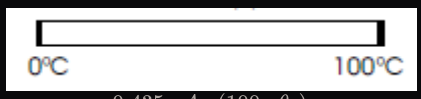
$1\mu F, 2\mu F$ **and** $3\mu F$ **are in parallel and hence, their equivalent capacitance is** $6\mu F$ **So, potential difference across** AB, BC, CD **will be** $4V$.

Q.23 Answer:

—

Solution:

The system constitute a compound rod (water + copper + water)



$\therefore Q = \frac{0.435 \times A \times (100-\theta_1)}{2\times10^{-3}}$ **.....(1)**

$= \frac{435 \times A \times (\theta_1-\theta_2)}{0.8}$ **.....(2)**

$= \frac{0.435\mathrm{A}(\theta_2-0)}{2\times10^{-3}}$ **.....(3)**

$\therefore \frac{100-\theta_1}{2} = \frac{\theta_1-\theta_2}{1} = \frac{\theta_2}{2}$

From (2) and $(3) 2\theta_1 = 3\theta_2$

From (1) and (3)

$2\theta_2 = 200 - 2\theta_1 = 200 - 3\theta_2$

$\therefore 5\theta_2 = 200 \Rightarrow \theta_2 = 40^\circ\mathrm{C}$

$\therefore$ **temperature gradient along the rod** $= \frac{60-40}{1} = 20^\circ\mathrm{Cm}^{-1}$ **and** $\theta_1 = 60^\circ\mathrm{C}$

Q.24 Answer:

—

Solution:

$$I_A = \frac{I_0}{2}$$

$$I_C = \frac{I_0}{2}\cos 30^\circ = \frac{I_0}{2}\frac{3}{4} = \frac{3I_0}{8}$$

$$I_B = I_C \cos^2 60^\circ = \frac{3I_0}{8}\cos^2 60^\circ = \frac{3}{32}I_0$$

$$\frac{I_B}{I_C} = \frac{3}{32} = 9.4\%$$

Q.25 Answer:

—

Solution:

$\because P = \frac{V^2}{R}$

$\therefore R = \frac{V^2}{P} = \frac{(200)^2}{100} = 400\Omega$

Resistance of half piece $= \frac{400}{2} = 200\Omega$

Resistance of pieces connected in parallel $= \frac{200}{2} = 100\Omega$

Energy liberated/second $= \mathrm{P} = \frac{\mathrm{V}^2}{\mathrm{R}} = \frac{200 \times 200}{100} = 400\ \mathrm{W}$

Q.26 Answer:

Vitamin B_{12}

Solution:

Pernicious anaemia is one of many types of the larger family of megaloblastic anaemia. It is caused by loss of gastric parietal cells, which are responsible, in part, for the secretion of intrinsic factor, a protein essential for subsequent absorption of vitamin B_{12} in the ileum.

Q.27 Answer:

Making it a cathode.

Solution:

Oxidation of iron takes place during the rusting.

It acts as an anode.

$\mathrm{Fe} \rightarrow \mathrm{Fe}^{+2} + 2\mathrm{e}^{-}$

Hence, coating or connecting iron with a metal of lower reduction potential (in comparison to iron) is the best way to prevent its rusting.

In this process, the metal with a low reduction potential undergoes oxidation and acts as an anode.

Therefore, iron acts as a cathode (cathodic protection).

The use of saline water accelerates the process of rusting, thereby, increasing the electric conduction of the electrolyte solution formed on the metal surface.

Q.28 Answer:

10 and 2

Solution:

For $n = 5, \ell = 0, 1, 2, 3, 4$

For every value of ℓ, $\mathrm{m}_\ell = 0$ **is possible.**

$\therefore$ **Total no. of electrons** $= 5 \times 2 = 10$

For $\mathrm{n} = 4, \ell = 0, 1, 2, 3$

Only for $\ell = 2$ **and** $\ell = 3,\ \mathrm{m}_\ell = +2$ **is possible**

$\therefore$ **Total no. of electrons (with** $\mathrm{m}_f = +2$ **)** $= 2 \times 2 = 4$

$\Rightarrow$ **No. of electrons** $\left(\right.$ **with** $\mathrm{m}_\mathrm{s} = -\frac{1}{2}\left.\right) = 2$

Q.29 Answer:

Colliision of atoms or molecules possessing sufficient threshold energy results into the product formation.

Solution:

According to the collision theory, the reactant molecules should possess minimum energy to change into the product. The minimum energy is called the threshold energy. It is not necessary that reactant molecules possess threshold energy. For obtaining the energy level of threshold energy, activation energy is given to the reactant molecules so that they can change into products.

Q.30 Answer:

$2 > 3 > 1 > 5 > 4$

Solution:

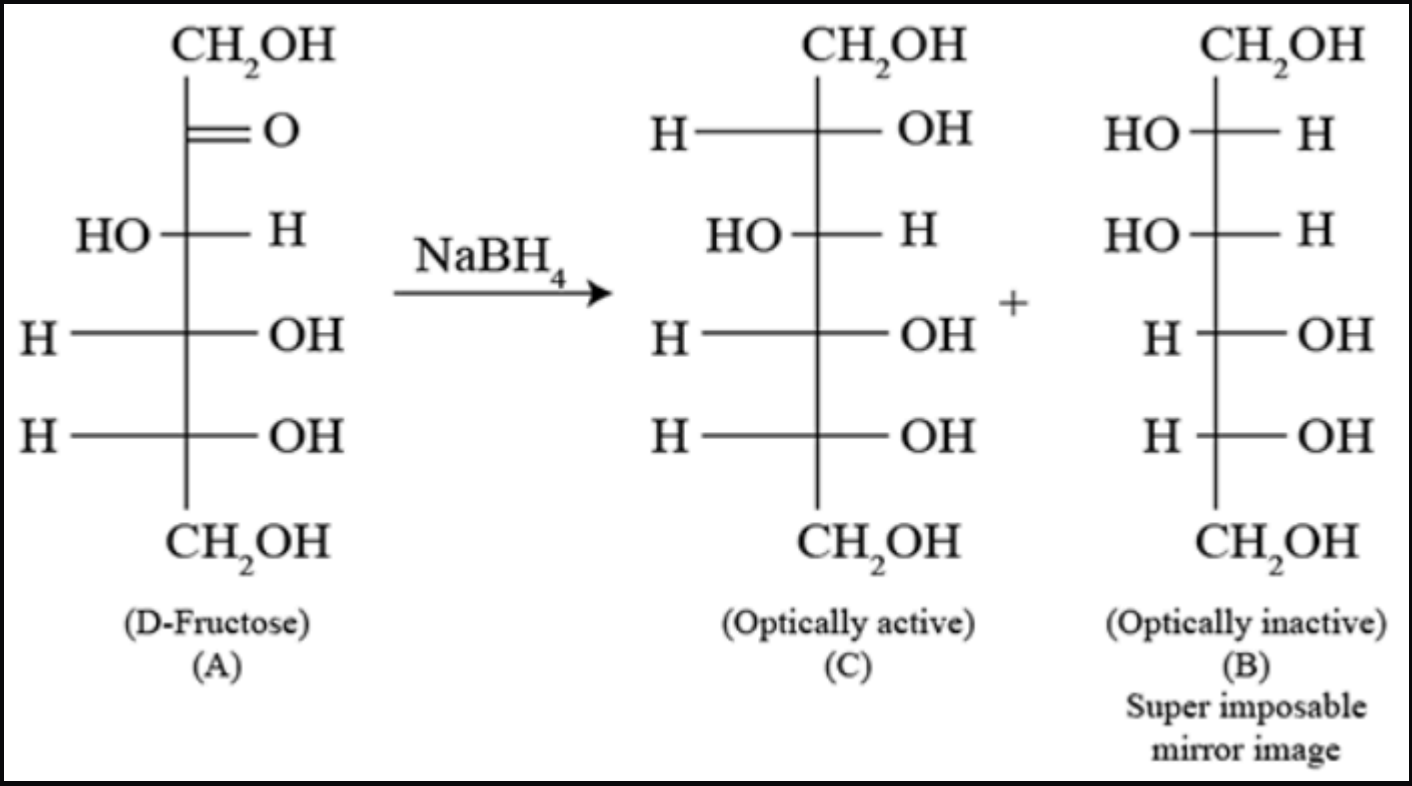
The rate of an electrophilic aromatic substitution reaction depends upon the stability of the arenium ion which is formed as an intermediate during the reaction. The group which stabilises the arenium ion gives the substitution reaction at a faster rate, while the group which decreases the stability, gives the reaction at a slower rate.

Among the given options, $-OCH_3$, $-C_2H_5$ increase the electron density while $-Br$, $-NO_2$ decrease the electron density, therefore the correct order is: $2 > 3 > 1 > 5 > 4$

Q.31 Answer:

D-fructose

Solution:



Q.32 Answer:

80 mL

Solution:

When phenolphthalein is used as an indicator

$0.05y = 10 \times 0.2 \times 1$

$y = 40 \text{ mL}$

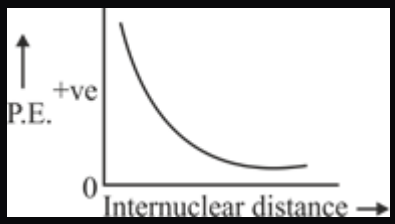
When methyl orange is used as an indicator.

$\therefore 0.05 \quad x = 10 \times 0.2 \times 3$

$x = 120 \text{ mL}$

Hence, $x - y = 80 \text{ mL}$

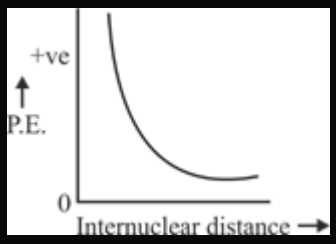
Q.33 Answer:



Solution:

As we know that, helium molecule has zero bond order and because of that, it can be said that they do not form bonds. And when bonds are formed, it only means that the stability increases and when helium atom's internuclear distance decreases, the molecular stability decreases and hence the potential energy increases.

The graph for such a condition is shown below:



Q.34 Answer:

If $\Delta G_{\text{system}} = 0$, the system has attained equilibrium

Solution:

If $\Delta G_{\text{system}} = 0$ **the system has attained equilibrium is right choice. In it alternative (d) is most confusing as when** $\Delta G > 0$**, the process may be spontaneous when it is coupled with a reaction which has** $\Delta G < 0$ **and total** ΔG **is negative, so right answer is (a).**

Q.35 Answer:

$\text{III} > \text{IV} > \text{II} > \text{I}$

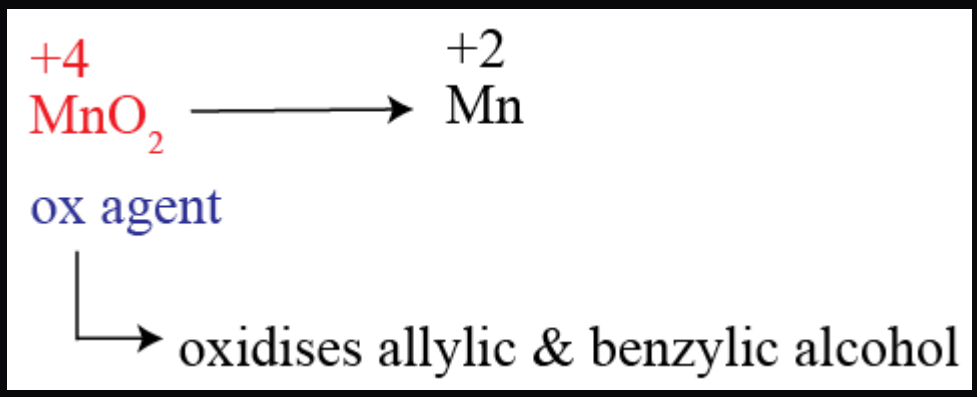
Solution:

Reactivity is directly proportional to stability of carbocation. Stability of carbocation formed is III > IV > II > I.

Q.36 Answer:

Hex-2-ene-1-ol

Solution:

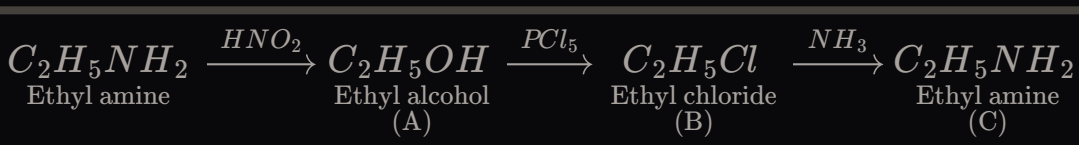


Only Hex-2-ene-1-ol is allylic

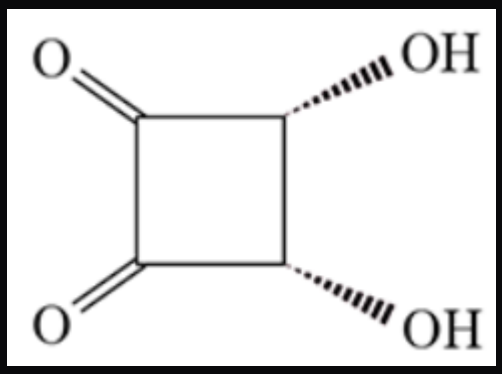
Q.37 Answer:

Ethyl amine

Solution:

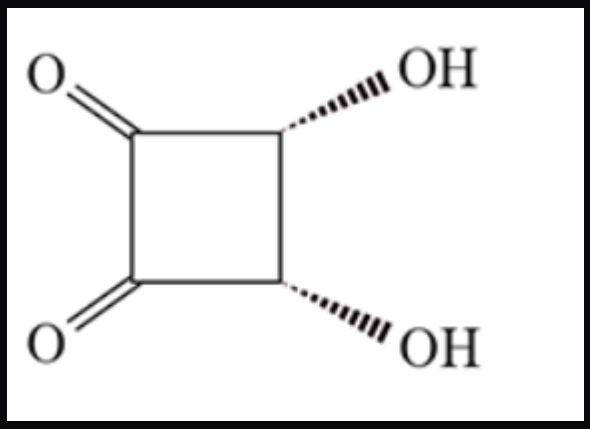


Q.38 Answer:

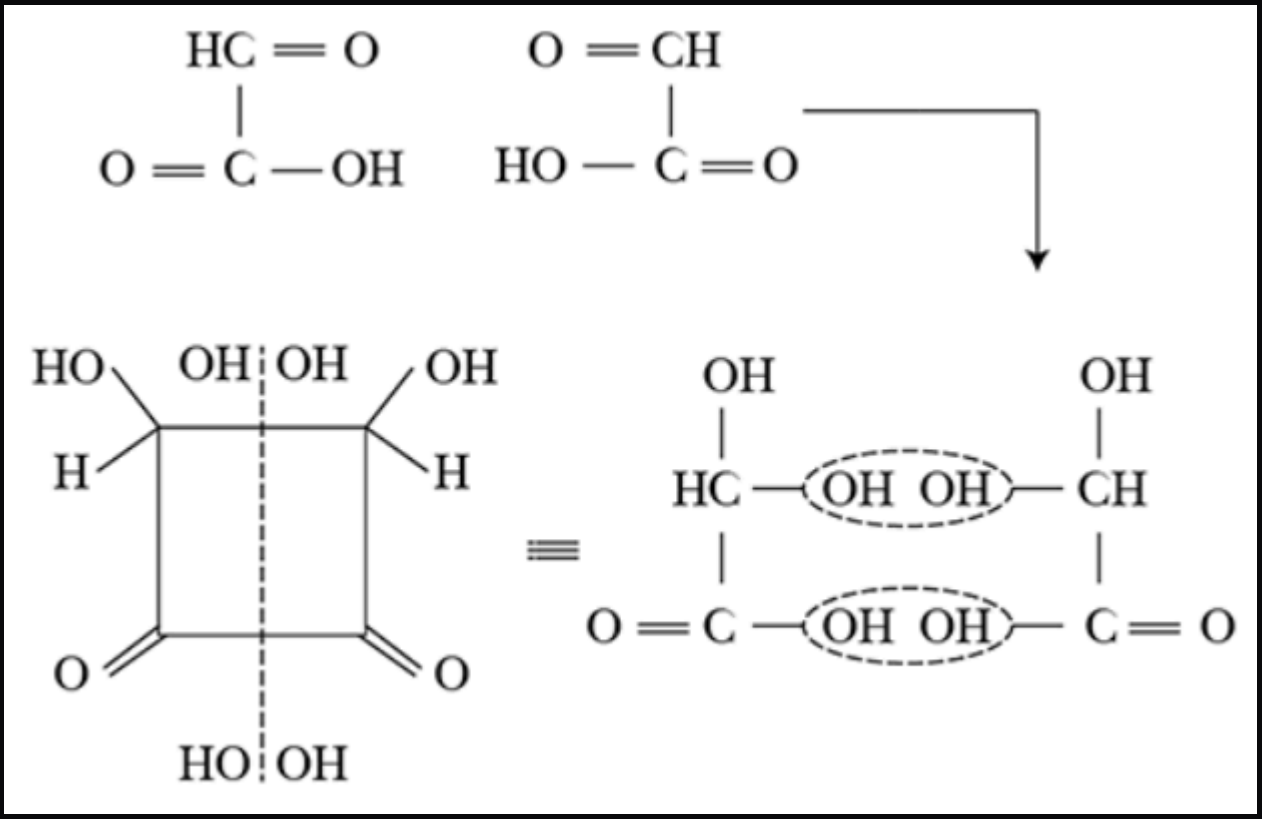


Solution:

Here the compound 'A' is



Glyoxalic acid is $(OHC-COOH)$. Proceed reverse

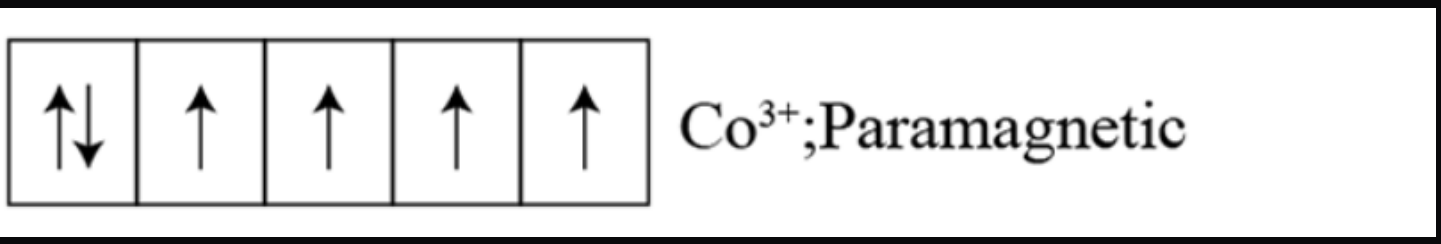


Q.39 Answer:

(ii), (iii) and (iv)

Solution:

Outer electronic configuration of Mn is $3\ d^5 4\ s^2$ which means it can show $+7$ oxidation state also (correct) Zinc does not form coloured ions as it has completely filled $3\ d^{10} 4\ s^2$ configuration. In $[CoF_6]^{3-}$, Co^{3+} is a d^6 system. Fluoride is a weak field ligand and hence it can not cause pairing of electrons.



Sc can show a maximum of $+3$ oxidation state as it has an outer electronic configuration of $3\ d^1 4\ s^2$. Zn exhibits only $+2$ oxidation state as this oxidation state is the most stable one for it. Hence (ii), (iii), (iv) are incorrect while (i), (v) are correct.

Q.40 Answer:

$8000\ cm^{-1}$

Solution:

The relation of CFSE for tetrahedral and octahedral complex is given as

$\Delta_t = \frac{4}{9}\Delta_0$

So Δ_t for $[CoCl_4]^{2-} = \frac{4}{9} \times 18000 cm^{-1}$

$= 8000\ cm^{-1}$.

Q.41 Answer:

C_6H_5CHO

Solution:

Aldehydes containing no α-hydrogen atom on warming with $50\%NaOH$ or KOH undergo disproportionation i.e. self oxidation - reduction known as Cannizzaro's reaction.

$$2C_6H_5CHO + NaOH \xrightarrow{50\%NaOH} C_6H_5COONa + C_6H_5CH_2OH$$

Q.42 Answer:

0.24

Solution:

In the saturated solution of sparingly soluble electrolyte, the ionic product of ions is constant at constant temperature and it is known a Solubility Product.

Here,

$Mg(OH)_2 \rightarrow [Mg^{2+}][OH^-]^2$

Given,

K_{sp} **for** $Mg(OH)_2 = 5.6 \times 10^{-12}$

$[Mg^{2+}] = 10^{-10}\ M$

Substituting the value, we get :

$K_{sp} = [Mg^{2+}][OH^-]^2 \Rightarrow 5.6 \times 10^{-12} = [10^{-10}][OH^-]^2 \Rightarrow OH^- = 0.24\ M$

Hence, the concentration of $[OH^-] = 0.24\ M$

Q.43 Answer:

Y exhibits variable covalency

Solution:

W: Phosphorus

Electronegativity : $O > Cl > S > P$

Electron Affinity : $Cl > S > O > P$

Y: Oxygen

Oxygen exhibits covalency of two only

X : Sulphur

Catenation : $S > P > O > Cl$

Z : Chlorine

Q.44 Answer:

Both (A) and (R) are true but (R) is not the correct explanation of (A)

Solution:

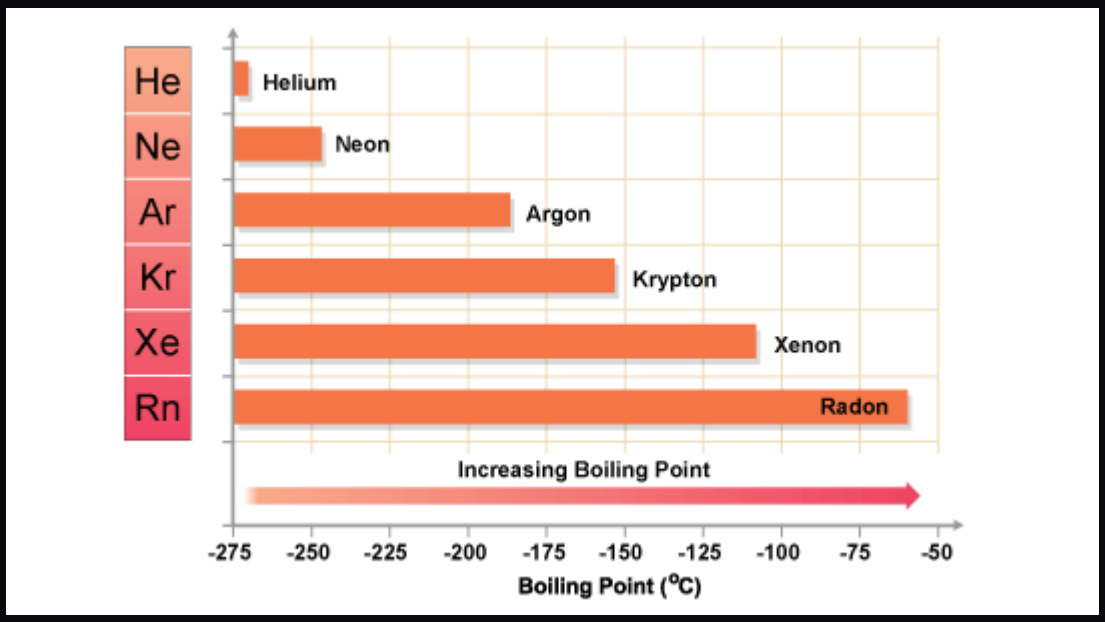
Both (A) **and** (R) **are true but** (R) **is not the correct explanation of** (A)

The melting and boiling points of noble gases are very low in comparison to those of other substances of comparable atomic and molecular masses. This indicates that only weak van der Waals forces or weak

London dispersion forces are present between the atoms of the noble gases in the liquid or the solid state. The van der Waals force increases with the increase in the size of the atom, and therefore, in general, the boiling and melting points increase from He to Rn.

Helium boils at −269 °C. Argon has larger mass than helium and have larger dispersion forces. Because of larger size the outer electrons are less tightly held in the larger atoms so that instantaneous dipoles are more easily induced resulting in greater interaction between argon atoms. Therefore, its boiling point (−186 °C) is more than that of He.

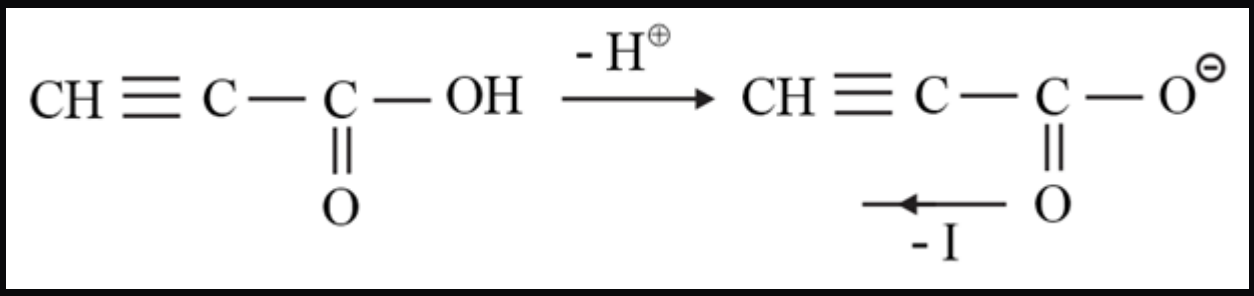
Similarly, because of increased dispersion forces, the boiling and melting points of monoatomic noble gases increase from helium to radon.

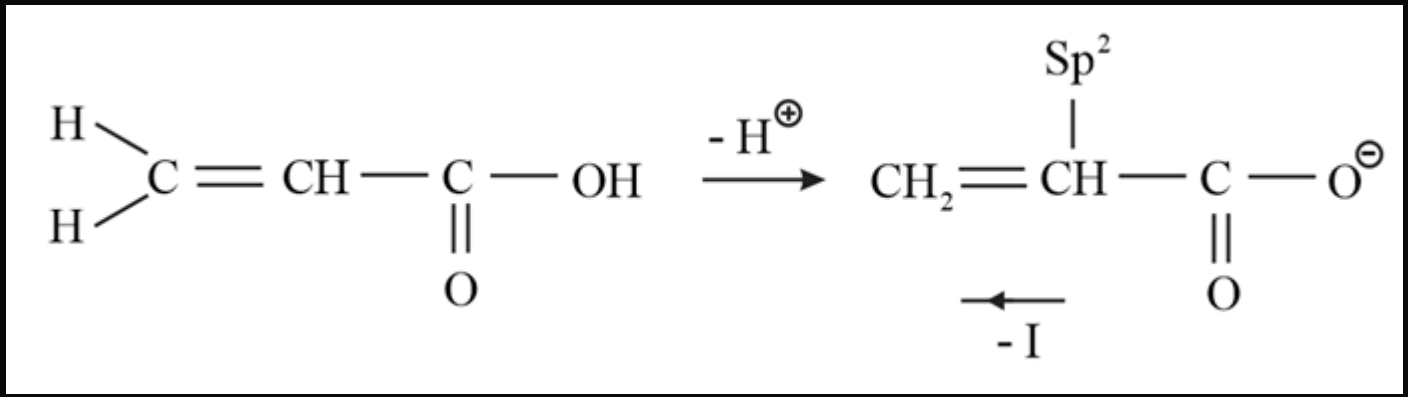


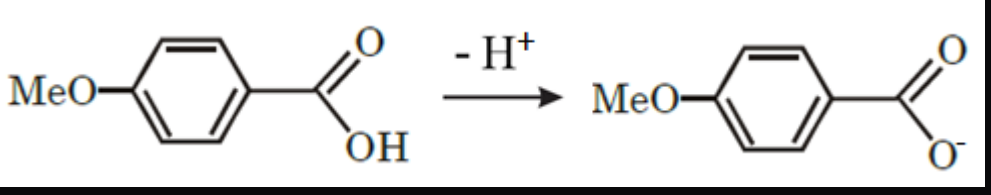
Q.45 Answer:

I > II > III > IV

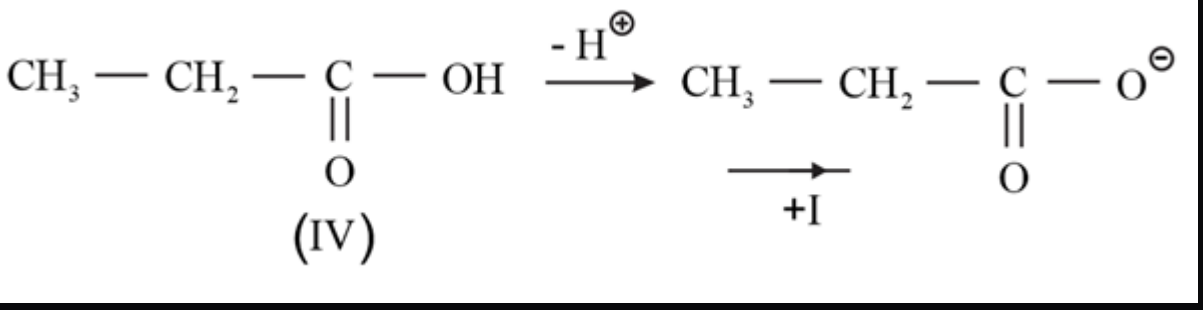
Solution:







The -OMe group is +M group, which decreases the acidic nature.



Acidic strength $\propto$ Stability of conjugate base

Order of Acidic strength $\Rightarrow$ $\boxed{\text{I} > \text{II} > \text{III} > \text{IV}}$

Q.46 Answer:

—

Solution:

$\Lambda^{\circ}_{m(C_6H_5COOH)} = \Lambda^{\circ}_{(C_6H_5COO^-)} + \Lambda^{o}_{(H^+)}$

$= 42 + 288.42 = 330.42$

$\alpha = \frac{\Lambda^{e}_{im}}{\Lambda^{m}_{m}} = \frac{12.8}{330.42} = 3.9\%$

Q.47 Answer:

—

Solution:

$NaHCO_3 + HCl \longrightarrow NaCl + H_2O + CO_2$

So moles of $NaHCO_3$(Pure)used = moles of HCl

$= 0.1 \times 5 \times 10^{-3}$

So weight of pure $NaHCO_3$ used $= 0.1 \times 5 \times$

$10^{-3} \times 84$ grams

$= 0.042$ grams

%Purity=$\frac{Mass\ of\ pure\ \ substance}{Total\ mass\ of\ sample}$

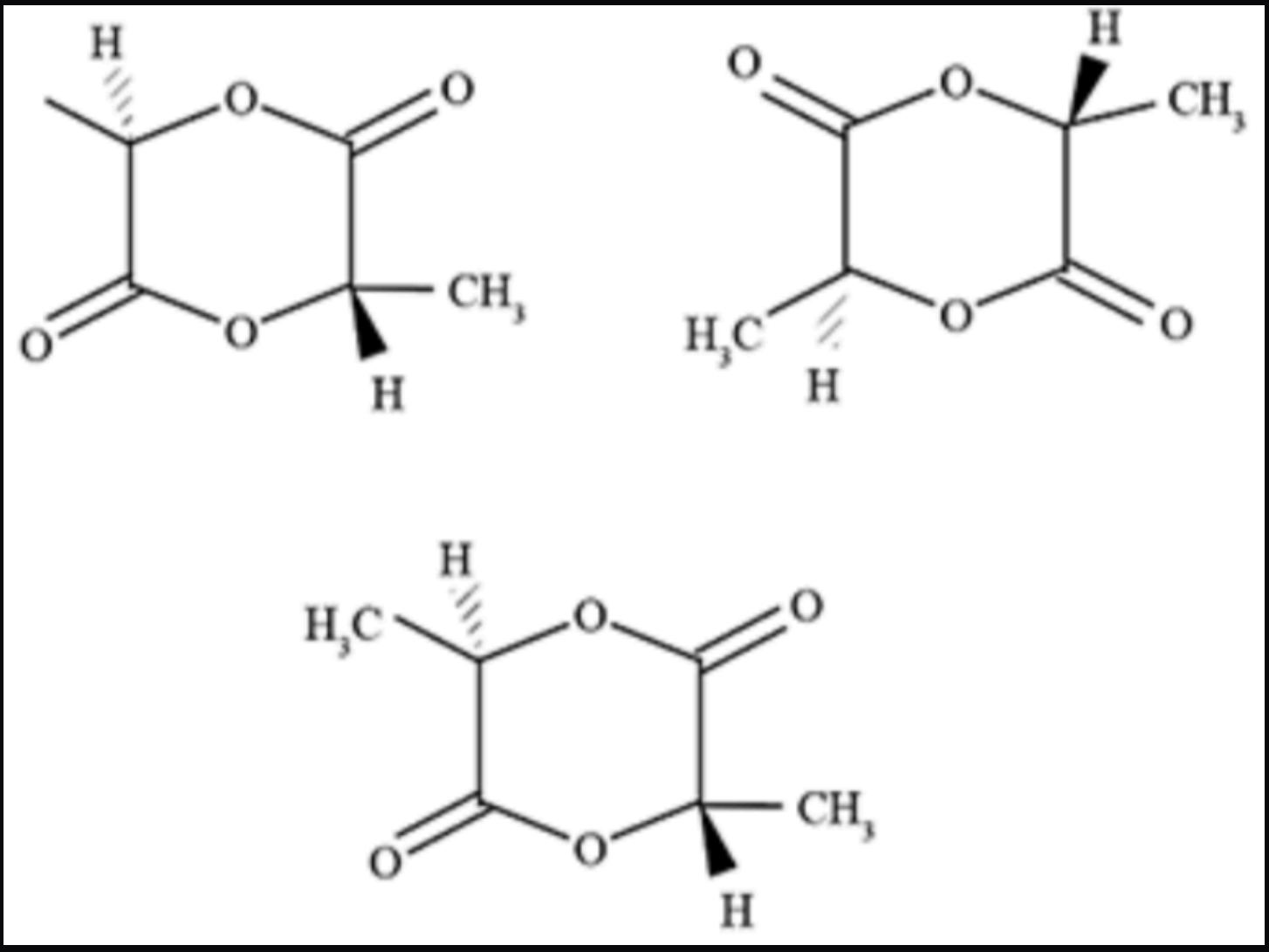
$\%$ Purity of tablet $= \frac{0.042}{0.14} \times 100 = 30\%$

Q.48 Answer:

—

Solution:

When lactic acid is heated, a cyclic diester derivatives are formed called lactides.



First and second isomers are optically active and third isomer have a point of symmetry. Hence, the correct answer will be 2 .

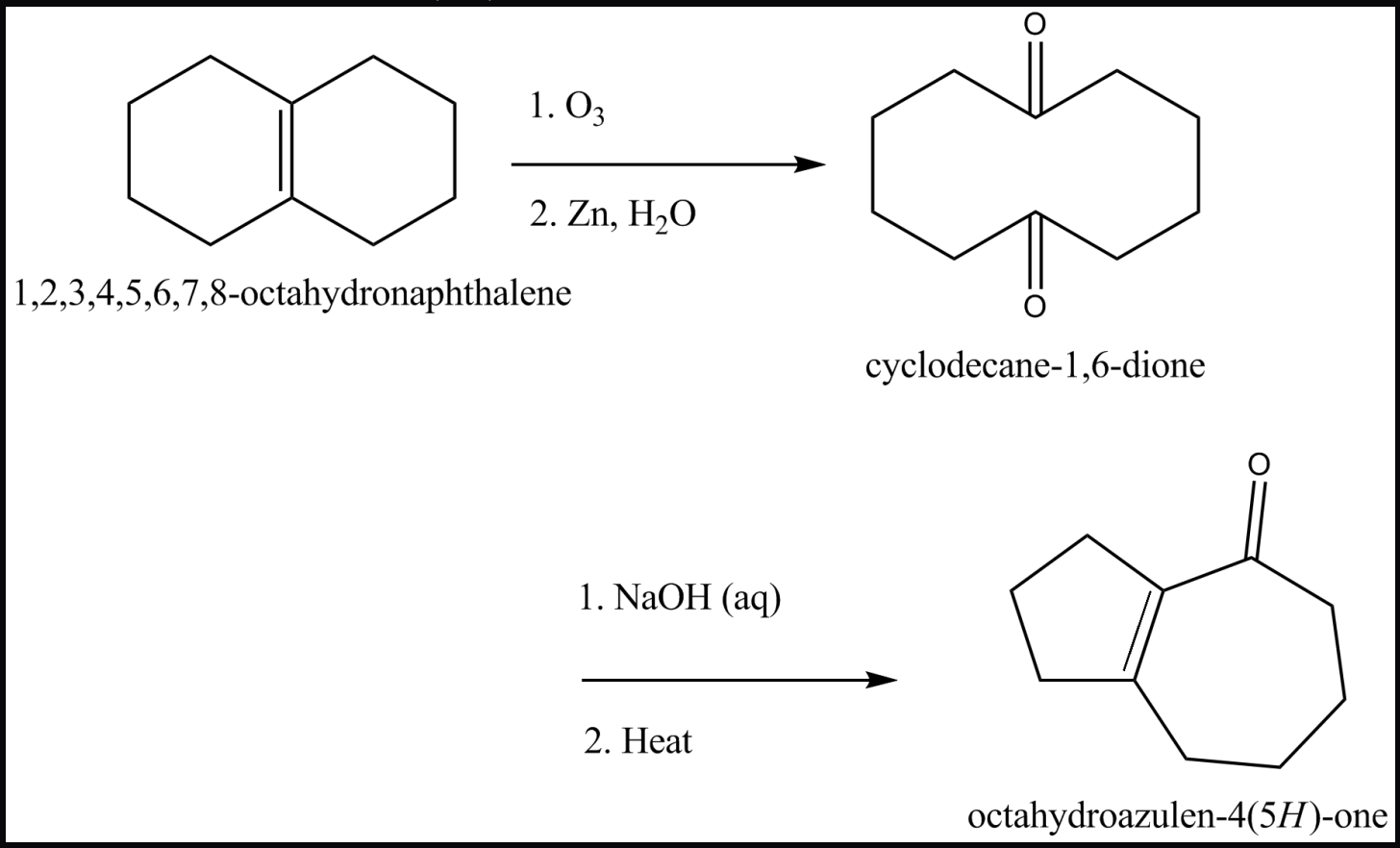
Q.49 Answer:

—

Solution:

Ozonolysis followed by reduction of $1,2,3,4,5,6,7,8$ octahydronaphthalene gives cyclodecane-1,6-dione. This on intramolecular aldol condensation gives only

one product octahydroazulen- $4(5\mathrm{H})$-one.



Q.50 Answer:

—

Solution:

	$CH_3O\,CH_{3(g)} \longrightarrow$	$CH_{4(g)} + H_{2(g)} + CO_{(g)}$		
at t = 0	0.4atm			
at t = 12min	0.4 − x	x	x	x

Now: $12 = \frac{2.303}{0.693} \times 14.5 \log(\frac{0.4}{0.4-\mathrm{x}})$

$0.25 = \log(\frac{0.4}{0.4-\mathrm{x}})$

$10^{0.25} = (\frac{0.4}{0.4-\mathrm{x}})$

$(\frac{0.4}{0.4-\mathrm{x}}) = 1.778$

$\Rightarrow \mathrm{x} = 0.175$

Total pressure $= (0.4 - 0.75) + 0.175 + 0.175 + 0.175 = 0.75$ atm

now: $\frac{\mathrm{x}}{100} = 0.75$

$\mathrm{x} = 75$

Q.51 Answer:

18

Solution:

$\vec{a} \cdot \vec{b} = x + y + z = 0$

If $x = 3, y = -3, z = 0$, **i.e.** $3! = 6$

If $x=6, y=-6, z=0$, i.e. $3!=6$
If $x=6, y=-3, z=-3$, i.e. $\frac{3!}{2!}=3$
If $x=-6, y=3, z=3$, i.e. $=\frac{3!}{2!}=3$
Total number of vectors $=18$

Q.52 Answer:

$g(x)$ is neither injective nor surjective

Solution:

Let remainder $g(x)$ be ax^3+bx^2+cx+d
$\because \quad f(x)=x^2\left(x^2-1\right)Q(x)+ax^3+bx^2+cx+d$, where $Q(x)$ is quotient
$\therefore$ RHS should have common factor x^2.
$\therefore \quad c=d=0$
$f(1)=a+b=3$
and $\quad f(-1)=-a+b=3$
$\therefore \quad b=3$ and $a=0$
$\therefore \quad g(x)=3x^2$ which is many one into function.
$\therefore \quad g(x)=0 \quad \Rightarrow \quad x=0$ lies between roots of $x^2-2(a+1)x+a(a-1)=0$
$\therefore \quad a(a-1)<0 \quad \Rightarrow \quad 0<a<1$

Q.53 Answer:

$\{-\pi, 0\}$

Solution:

$\left[x^2-\frac{1}{3}\right]=\left[x^2+\frac{2}{3}-1\right]=\left[x^2+\frac{2}{3}\right]-1=k(\text{ let })$
Now, the function is $\sin^{-1}k-\cos^{-1}(k+1)$
For the above function to be defined, $-1 \leq k \leq 1 \& -1 \leq (k+1) \leq 1$
$\therefore k=\{-1,0\}$
$\therefore$ Range $=\left\{\sin^{-1}(-1)-\cos^{-1}(0), \sin^{-1}(0)-\cos^{-1}(1)\right\}=\{-\pi, 0\}$

Q.54 Answer:

7

Solution:

We have, $\sum_{i=1}^{n}\left(x_i+1\right)^2=11n \ldots \ldots .$ (i)
and $\sum_{i=1}^{n}\left(x_i-1\right)^2=7n \ldots$ (ii)
Adding (i) and (ii) , we get
$2\sum_{i=1}^{n}\left(x_i^2+1\right)=18n \Rightarrow \sum_{i=1}^{n}\left(x_i^2+1\right)=9n$
$\Rightarrow \sum_{i=1}^{n}x_i^2+n=9n \Rightarrow \sum_{i=1}^{n}x_i^2=8n$
$\Rightarrow \frac{\sum_{i=1}^{n}x_i^2}{n}=8$
Subtracting (i) and (ii), we get

$4\sum_{i=1}^{n} x_i = 4n \Rightarrow \sum_{i=1}^{n} x_i = n \Rightarrow \frac{\sum_{i=1}^{n} x_i}{n} = 1$

Now, variance $= \frac{1}{n}\left[\sum_{i=1}^{n} x_i^2\right] - \left[\frac{\sum_{i=1}^{n} x_i}{n}\right]^2 = 8 - 1 = 7$

Q.55 Answer:

$\frac{3}{4}$

Solution:

Given equation is $\frac{1}{\sqrt{y}}\frac{dy}{dx} + \frac{x}{1-x^2}\sqrt{y} = x$

Let, $2\sqrt{y} = \nu$

$\Rightarrow \frac{1}{\sqrt{y}}\frac{dy}{dx} = \frac{d\nu}{dx}$

Thus, we have $\frac{d\nu}{dx} + \frac{x}{2(1-x^2)}\nu = x$

$\therefore \text{I.F.} = e^{\int \frac{x}{2(1-x^2)}dx}$

$= e^{\int \frac{-d\left(1-z^2\right)}{\sin\left(1-x^2\right)}}$

$= e^{-\frac{1}{4}\ln\left(1-x^2\right)}$

$= \left(1-x^2\right)^{-\frac{1}{4}}$

Thus, the solution is $\nu\left(1-x^2\right)^{-\frac{1}{4}} = \int x\left(1-x^2\right)^{-\frac{1}{4}}dx$

or $\nu \cdot \left(1-x^2\right)^{-\frac{1}{4}} = -\frac{2}{3}\left(1-x^2\right)^{\frac{3}{4}} + C'$

or $2\sqrt{y} = -\frac{2}{3}\left(1-x^2\right) + C'\left(1-x^2\right)^{\frac{1}{4}}$

$\Rightarrow \sqrt{y} = -\frac{\left(1-x^2\right)}{3} + C\left(1-x^2\right)^{\frac{1}{4}}$

$\Rightarrow f(x) = 1 - x^2$

$\Rightarrow f\left(-\frac{1}{2}\right) = 1 - \frac{1}{4} = \frac{3}{4}$

Q.56 Answer:

$3:1$

Solution:

Let position vectors of O, A, B, C **are** $\vec{O}, \vec{a}, \vec{b}, \vec{c}$ **respectively.**

$2OA + 3OB + 4OC = 0$

$\Rightarrow 2\vec{a} + 3\vec{b} + 4\vec{c} = 0$

Area of $\Delta AOC = \frac{1}{2}|\overrightarrow{OA} \times \overrightarrow{OC}| = \frac{1}{2}|\vec{a} \times \vec{c}|$

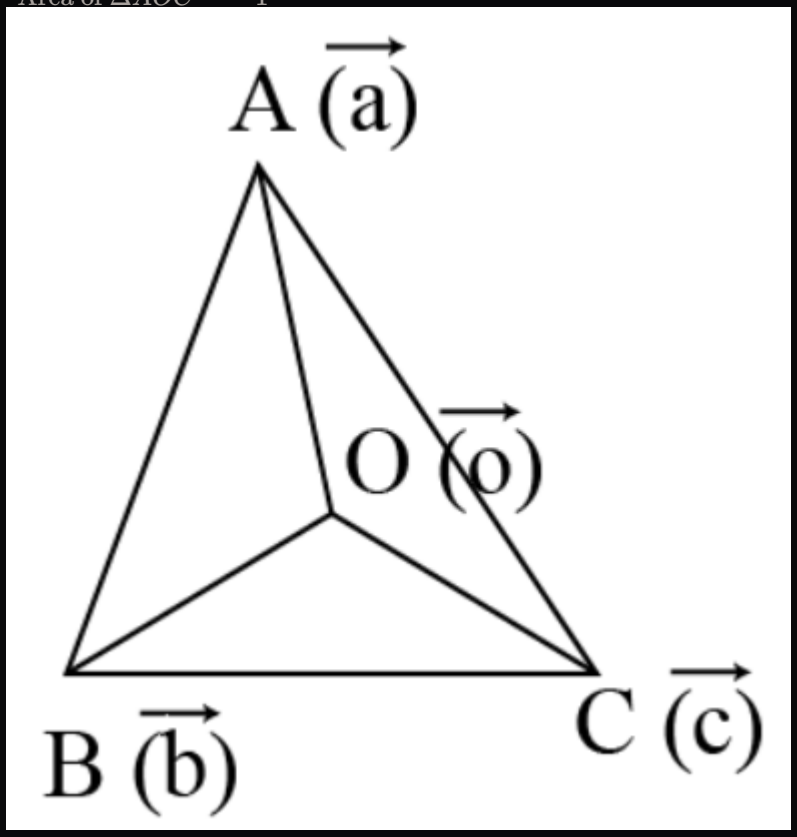
Area of $\Delta ABC = \frac{1}{2}|\overrightarrow{BA} \times \overrightarrow{BC}| = \frac{1}{2}|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|$

$= \frac{1}{2}\mid \vec{a} \times \left(\frac{-4\vec{c}-2\vec{a}}{3}\right) + \left(\frac{-4\vec{c}-2\vec{a}}{3}\right) \times \vec{c} + \vec{c} \times \vec{a}$

$= \frac{1}{2}\left|\frac{4}{3}\vec{c} \times \vec{a} + \frac{2}{3}\vec{c} \times \vec{a} + \vec{c} \times \vec{a}\right|$

$= \frac{3}{2}|\vec{c} \times \vec{a}|$

$\frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta AOC} = \frac{3}{1}$



Q.57 Answer:

$1 - \frac{1}{\sqrt{1-x^2}}$

Solution:

We can reduce the function by substituting

$x = \cos\theta \Rightarrow \theta = \cos^{-1}x$

$y = \cos\left(2\tan^{-1}\left(\sqrt{\frac{1-\cos\theta}{1+\cos\theta}}\right)\right) - 2\cos^{-1}\left(\sqrt{\frac{1-\cos\theta}{2}}\right)$

$y = \cos\left(2\tan^{-1}\left(\sqrt{\frac{2\sin^2\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}}}\right)\right) - 2\cos^{-1}\left(\sqrt{\frac{2\sin^2\frac{\theta}{2}}{2}}\right)$

$= \cos\left(2\tan^{-1}\left(\tan\frac{\theta}{2}\right)\right) - 2\cos^{-1}\left(\sin\frac{\theta}{2}\right)$

As $\theta = \cos^{-1}x$

$\theta \in [0, \pi) \Rightarrow \frac{\theta}{2} \in \left[0, \frac{\pi}{2}\right)$

In the first quadrant $\tan^{-1}\left(\tan\frac{\theta}{2}\right) = \frac{\theta}{2}$

$= \cos\left(2 \times \frac{\theta}{2}\right) - 2\cos^{-1}\left(\sin\frac{\theta}{2}\right)$

As $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$

$y = \cos\theta - 2\left(\frac{\pi}{2} - \sin^{-1}\left(\sin\frac{\theta}{2}\right)\right)$

As $\sin^{-1}\left(\sin\frac{\theta}{2}\right) = \frac{\theta}{2}$ **in the first quadrant**

$y = \cos\theta - \pi + \theta$

$y = x - \pi + \cos^{-1}x$

$\frac{dy}{dx} = 1 - \frac{1}{\sqrt{1-x^2}}$

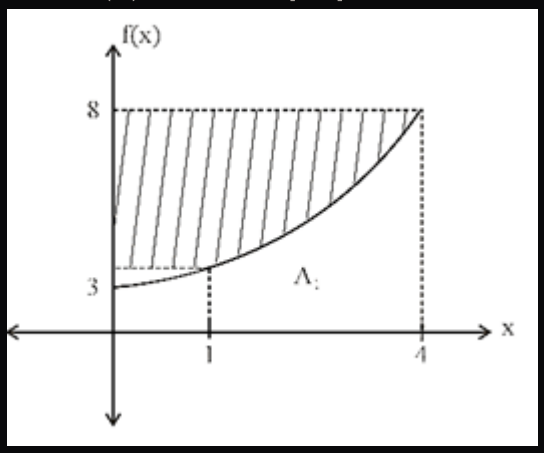
Q.58 Answer:

17

Solution:

Since $f(x) > 0 \ \forall x \in [1,4]$

$\Rightarrow f^{-1}(x) > 0 \ \forall x \in [3,8]$



Note that the shaded area is the required area and $A_1 = 12$.

Let the required area be A_2

Also, $A_1 + A_2 = 32 - 3 = 29$

$\Rightarrow A_2 = 17$ sq. units

Q.59 Answer:

c, a and b are in A.P.

Solution:

$4\left(9c^2 + b^2\right) + 9\left(a^2 - 2ac\right) = 6b(2c + a)$

$\Rightarrow (3a)^2 + (2b)^2 + (6c)^2 - (3a)(2b) - (2b)(6c) - (6c)(3a) = 0 \quad \Rightarrow 3a = 2b = 6c \quad \Rightarrow \frac{a}{2} = \frac{b}{3} = c = \lambda$

$\Rightarrow a = 2\lambda \quad b = 3\lambda \quad c = \lambda$

So, c, a, b, are in A.P.

Q.60 Answer:

$\frac{1}{12}$

Solution:

Winning the series with margin of 4 wins or more can be done when series finishes with a scoreline of $4 - 0$ or $5 - 0$ in favour of India.

$\Rightarrow 5 \times \left(\frac{1}{2^4}\right) \times \left(\frac{1}{6}\right) + \left(\frac{1}{2^5}\right) = \frac{1}{12}$

Q.61 Answer:

9

Solution:

Solution: We know $x^2 + y^2 + z^2 = 5$.

$$\begin{aligned} \therefore \quad 5 = x^2 + y^2 + z^2 &= (x + y + z)^2 - 2xy - 2yz - 2xz \\ &= 9 - 2(xy + yz + xz) \\ \Rightarrow \quad xy + yz + zx &= 2 \end{aligned}$$

We know that

$x^3+y^3+z^3-3xyz=(x+y+z)\left[x^2+y^2+z^2-(xy+yz+xz)\right]$

$\Rightarrow 7-3xyz=3[5-2]=9$

$\Rightarrow 7-3xyz=3[5-2]=9$

Or, $xyz=\frac{-2}{3}$..

$x^4+y^4+z^4=\left(x^2+y^2+z^2\right)^2-2\left[(xy)^2+(yz)^2+(zx)^2\right]$

$=25-2\left[(xy+yz+zx)^2-2\left(xy^2z+yz^2x+zx^2y\right)\right]$

$=25-2\left[2^2-2xyz(x+y+z)\right]$

$=25-2\left[4+\frac{4}{3}\times 3\right]$

$=25-16=9$

Q.62 Answer:

$a\geq\sqrt{2}$

Solution:

We have; $\mathrm{f(x)=\sin x-\cos x-ax+b}$

$\Rightarrow f'(x)=\cos x+\sin x-a$

$\Rightarrow \mathrm{f'(x)<0\forall x\in R}$

$\Rightarrow (\cos x+\sin x)$ As the max. value of $(\cos x+\sin x)$ is $\sqrt{2}$

The above is possible when $a\geq\sqrt{2}$

Q.63 Answer:

symmetric but not transitive and reflexive

Solution:

$\mathrm{|x|+|y|\leq\frac{1}{2}}$

If $\mathrm{x=y,\quad 2|x|\leq\frac{1}{2},-\frac{1}{4}\leq x\leq\frac{1}{4}}$

$\therefore \mathrm{(x,x)\notin R}$ $\quad$ vx $\in$ real no.

$\therefore \mathrm{R}$ is not reflexive

If $|x|+|y|\leq\frac{1}{2}\quad\Rightarrow|y|+|x|\leq\frac{1}{2}$

$\therefore\quad (x,y)\in R\quad\Rightarrow\quad (y,x)\in R$

$\therefore \mathrm{R}$ is symmetric

If $|x|+|y|\leq\frac{1}{2}$ and $|y|+|z|\leq\frac{1}{2}\Rightarrow|x|+|z|\leq\frac{1}{2}$

$\therefore \mathrm{R}$ is not transitive.

Q.64 Answer:

17

Solution:

$5! = 120 \Rightarrow (5!)^2 = 12^2 \times 100$

$\Rightarrow (5!)^2, (6!)^2, (7!)^2, \ldots\ldots, (99!)^2$ all are divisible by 100

and $(1!)^2 + (2!)^2 + (3!)^2 + (4!)^2$

$= 1 + 4 + 36 + 576 = 617$

So, when divided by 100 , the remainder obtained will be 17 .

Q.65 Answer:

(0, c, c)

Solution:

Let L_1, L_2, L_3 be the given mutually perpendicular lines and P(x_0, y_0, z_0) be their point of concurrency. Let us, also assume that L_1 cuts the x-axis at A(a, 0, 0), L_2 meets the y-axis at B(0, b, 0) and L_3 passes through C(0, 0, c) (where a and b are variables and c is a constant), then

L_1 parallel to the line with DR's $(x_0 - a, y_0, z_0)$

L_2 parallel to the line with DR's $(x_0 , y_0 - b , z_0)$

and

L_3 parallel to the line with DR's $(x_0 , y_0 , z_0 - c)$

Hence, we can say that

$$\left.\begin{array}{ll} L_1 \perp L_2 & \Rightarrow \quad x_0(x_0 - a) + y_0 (y_0 - b) + z_0^2 = 0 \\ L_2 \perp L_3 & \Rightarrow \quad x_0^2 + (y_0 - b)y_0 + z_0(z_0 - c) = 0 \\ L_3 \perp L_1 & \Rightarrow \quad x_0(x_0 - a) + y_0^2 + z_0(z_0 - c) = 0 \end{array}\right\}$$

$$\Rightarrow \left.\begin{array}{ll} x_0^2 + y_0^2 + z_0^2 - ax_0 - by_0 = 0 & \ldots\ldots\ldots (i) \\ x_0^2 + y_0^2 + z_0^2 - cz_0 - by_0 = 0 & \ldots\ldots\ldots (ii) \\ x_0^2 + y_0^2 + z_0^2 - ax_0 - cz_0 = 0 & \ldots\ldots\ldots (iii) \end{array}\right\}$$

Eliminating a and b from the above equations (by applying the operation Eq. (i) - Eq. (ii) - Eq. (iii), we get

$x_0^2 + y_0^2 + z_0^2 - 2cz_0 = 0$

Hence, the required locus of the point P is given by $x^2 + y^2 + z^2 - 2cz = 0$

Options (0, c, c) satisfies the above equation.

Q.66 Answer:

36

Solution:

$$A = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$$

$|A| = abc = 120$

$\Rightarrow (a, b, c)$

$1, 1, 120 \Rightarrow 3!/2! \quad 2, 2, 3 \Rightarrow 3!/2! \quad 3, 4, 10 \Rightarrow 3!$

$1, 2, 60 \Rightarrow 3! \quad 2, 3, 20 \Rightarrow 3! \quad 3, 5, 8 \Rightarrow 3!$

$1, 3, 40 \Rightarrow 3! \quad 2, 4, 15 \Rightarrow 3! \quad 3, 8, 5 \Rightarrow 3!$

$1, 4, 30 \Rightarrow 3! \quad 2, 6, 10 \Rightarrow 3! \quad 4, 5, 6 \Rightarrow 3!$

$1, 6, 20 \Rightarrow 3!$

$1, 8, 15 \Rightarrow 3!$

$1, 10, 12 \Rightarrow 3!$

Case 1 :(a,b,c are positive) $\Rightarrow 14(3!) + 2(3!/2!)$

$= 14 \times 6 + 2 \times 3$

$= 90$

Case 2: (any two are –ve): ${}^{3}\mathrm{C}_{2}{}^{1}\mathrm{C}_{1}(90)$

$= 3 \times 1 \times 90$

$= 270$

Total matrices $= 90 + 270 = 10\mathrm{n}$

$= 360 = 10n$

$= n = 36$

Q.67 Answer:

$\frac{x^2}{16} - \frac{y^2}{9} = -1$

Solution:

The eccentricity of $\frac{x^2}{16} + \frac{y^2}{25} = 1$ **is**

$e_1 = \sqrt{1 - \frac{16}{25}} = \frac{3}{5}$

$\therefore e_2 = \frac{5}{3} \quad (\because e_1 e_2 = 1)$

Focus of the Ellipse $(0, \pm 5)$

Hence, the equation of the Hyperbola is

$\frac{x^2}{16} - \frac{y^2}{9} = -1.$

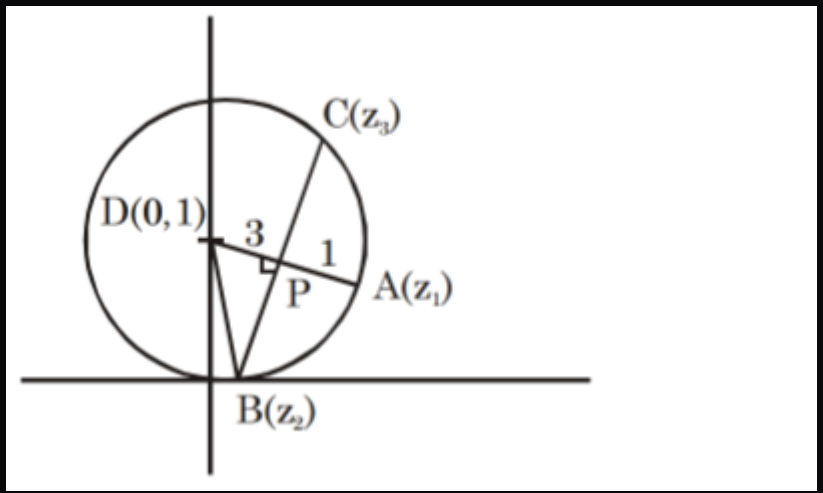
Q.68 Answer:

$\frac{\sqrt{7}}{4}$

Solution:

$$\frac{3z_1 + \mathrm{i}}{4} = \frac{2z_2 + 2z_3}{4}$$

$\mathrm{P} =$ point of intersection of AD&BC



$\mathrm{AD} = 1 \Rightarrow \mathrm{DP} = \frac{3}{4}$

$\mathrm{BP} = \sqrt{1 - \frac{9}{16}} = \frac{\sqrt{7}}{4} \Rightarrow \mathrm{BC} = \frac{\sqrt{7}}{2}$

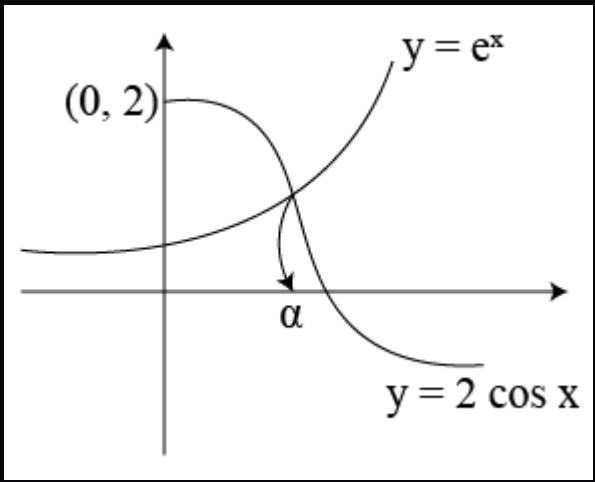
$\text{area of Quad. ABCD} = \frac{1}{2} \cdot \mathrm{AD} \times \mathrm{BC}$

$= \frac{1}{2} \times 1 \times \frac{\sqrt{7}}{2} = \frac{\sqrt{7}}{4}$

Q.69 Answer:

$I_1 > I_2$

Solution:



As, $x \in (0, \alpha) \Rightarrow 2\cos x > e^x$

$\therefore \frac{1+2\cos x}{1+e^x} > \frac{1+e^x}{1+2\cos x}$

$\Rightarrow \int_0^\alpha \frac{1+2\cos x}{1+e^x} dx > \int_0^\alpha \frac{1+e^x}{1+2\cos x} dx$

$\Rightarrow I_1 > I_2$

Also, $I_1, I_2 > 0$

Q.70 Answer:

-1

Solution:

The matrix form of the given system of equation is:

$$\begin{bmatrix} k & 2 & -1 \\ 0 & k-1 & -2 \\ 0 & 0 & k+2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Therefore, the coefficient matrix will have a solution iff it's determinant $\neq 0$

i.e. $(\mathrm{k}-1)(\mathrm{k}+2) \neq 0$

i.e., $\mathrm{k} \neq 0, 1, -2$

Thus the given system of equation will have a solution only at $\mathrm{k} = -1$.

Q.71 Answer:

—

Solution:

The polynomial is an everywhere differentiable function. Therefore, the points of extremum can only be roots of derivative. Further more, the derivative of a polynomial is a polynomial.

The polynomial of the least degree with roots $x_1 = 1$ and $x_2 = 3$ has the form $a(x-1)(x-3)$.

Hence, $P'(x) = a(x-1)(x-3) = a\left(x^2 - 4x + 3\right)$

Since, at $x = 1$, there must be $P(1) = 6$, we have

$$P(x) = \int_1^x P'(x)dx + 6 = a\int_1^x \left(x^2 - 4x + 3\right)dx + 6$$

$$= a\left(\frac{x^3}{3} - 2x^2 + 3x - \frac{4}{3}\right) + 6$$

Also, $\quad P(3) = 2 \Rightarrow a = 3$

Hence, $\quad P(x) = x^3 - 6x^2 + 9x + 2$

Now, $\quad P(2) = 8 - 24 + 18 + 2 = 4$, also

$P'(x) = 3\left(x^2 - 4x + 3\right) \Rightarrow P'(0) = 9$

$\therefore \quad P(2) + P'(0) - 7 = 6$

Q.72 Answer:

—

Solution:

Three digits occur twice each

$\frac{6!}{2!2!2!} \times 10 = 900$

Two digits occur three each

$\frac{6!}{3!3!} \times 10 = 200$

One digit occurs four times and the other twice $2 \times \frac{6!}{4!2!} \times 10 = 300$

Finally all digits are the same. There are 5 such numbers.

Thus the total number of admissible numbers is $900 + 200 + 300 + 5 = 1405$

Q.73 Answer:

—

Solution:

The extremities

Since circle circumscribes the rectangle and so its diagonal with vertices $(-4, 4)$ and $(6, -1)$ will act as diameter of the circle. Hence its equation is

$$(x+4)(x-6) + (y-4)(y+1) = 0$$

$$\Rightarrow x^2 + y^2 - 2x - 3y - 28 = 0$$

Length of y intercept is $AB = 11$

Q.74 Answer:

—

Solution:

$\because x, y, z \in N$ and

$x \in A$, i.e., $1 \leq x \leq 8$

and $y \in B$, i.e., $\quad 9 \leq y \leq 16$

and $z \in C$, i.e., $\quad 17 \leq z \leq 24$

$\therefore$ Number of solution for $x + y + z = 36$ is equal to coefficient of x^{36} in
$\left(x + x^2 + \ldots + x^8\right) . \left(x^9 + x^{10} + \ldots + \ x^{16}\right) \cdot \left(x^{17} + x^{18} + \ldots + x^{24}\right)$

$=$ coefficient of x^9 in $\left(1 + x + \ldots + x^7\right)^3$

$=$ coefficient of x^9 in $\frac{\left(1-x^8\right)^3}{(1-x)^3}$

$=$ coefficient of x^9 in $\left(1 - 3x^8 + 3x^{16} - x^{24}\right)$

$\left(1 + 3x + \frac{3.4}{2!}x^2 + \ldots\right)$

$= {}^{11}C_9 - 3.3$

$= 55 - 9$

$= 46$

Q.75 Answer:

—

Solution:

$2h = a\left(t_1^2 + t_2^2\right) \ldots (1)$
and $2\mathrm{k} = 2\mathrm{a}\left(\mathrm{t}_1 + \mathrm{t}_2\right) \ldots (2)$
and $t_1 t_2 = -1$
from (2) $k^2 = a^2\left[\frac{2h}{a} - 2\right]$
$y^2 = 2a(x - 2a)$
Put $a = 2$

Hence, $y^2 = 4(x-4)$.

